

A Bi-Hemisphere Domain Adversarial Neural Network Model for EEG Emotion Recognition

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Abstract—In this paper, we propose a novel neural network model, called bi-hemisphere domain adversarial neural network (BiDANN) model, for electroencephalograph (EEG) emotion recognition. The BiDANN model is inspired by the neuroscience findings that the left and right hemispheres of human's brain are asymmetric to the emotional response. It contains a global and two local domain discriminators that work adversarially with a classifier to learn discriminative emotional features for each hemisphere. At the same time, it tries to reduce the possible domain differences in each hemisphere between the source and target domains so as to improve the generality of the recognition model. In addition, we also propose an improved version of BiDANN, denoted by BiDANN-S, for subject-independent EEG emotion recognition problem by lowering the influences of the personal information of subjects to the EEG emotion recognition. Extensive experiments on the SEED database are conducted to evaluate the performance of both BiDANN and BiDANN-S. The experimental results have shown that the proposed BiDANN and BiDANN models achieve state-of-the-art performance in the EEG emotion recognition.

Index Terms—EEG emotion recognition, long short term memory (LSTM), cerebral hemisphere asymmetry, adversarial network

1 INTRODUCTION

EMOTION, as a common mental phenomenon, is closely related to human's cognition and behavior [1]. Although emotion can be easily captured by human beings, it is still hard to be understood by machines. As one of the most active research topics of affective computing [2], emotion recognition had received substantial attention from computer vision and pattern recognition research communities. Basically, the responses of emotion can be roughly divided into the external and the internal responses. The typical external responses include facial expression, gesture or speech of human beings, and the typical internal responses include skin conductance response, heart rate, blood pressure, respiration rate, electroencephalograph (EEG), magnetoencephalogram (MEG) [3]. From the neuroscience view of point [4],

there are some major brain cortex regions, e.g., the orbital frontal cortex, ventral medial prefrontal cortex, and amygdala, are closely related to emotions [5], [6], [7], which provides us a potential way to decode emotion by recording human's brain signals over these brain regions. For example, by placing the EEG electrodes on the scalp, we can record the neural activities of the brain, which can be used to recognize human's emotions.

Traditional EEG emotion recognition system usually consists of two major parts, i.e., feature extraction part and classifier design part. EEG features can be extracted either from time domain, frequency domain, or from time-frequency domain [8]. Jenke et al. [8] provided a comprehensive survey on the EEG feature extraction approaches. For dealing with the classification problem, many EEG emotion recognition models and methods were proposed over the past several years [9], [10]. Zheng et al. [11] proposed a novel group sparse canonical correlation analysis (GSCCA) method for simultaneous EEG channel selection and emotion recognition. Li et al. [12] proposed a graph regularized sparse linear regression (GRSLR) method to deal with EEG emotion recognition problem. Recently, using deep learning methods for EEG emotion recognition had been widely adopted and had demonstrated better performance than traditional methods. In [13], Zheng et al. proposed to use Deep Belief Network (DBN) for EEG emotion classification. In [14], Song et al. used a graph to model the multichannel EEG features and then perform EEG emotion classification based on it.

Although many algorithms or models have been proposed for the EEG emotion recognition problem, most of them focused on the scenarios where both training and testing data come from the same domain, in which we usually assume that the feature distributions of training and testing data are

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一个用于脑电图情绪识别的双半球域对抗神经网络模型

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摘要—本文提出了一种新型神经网络模型，称为双半球域对抗神经网络（BiDANN）模型，用于脑电图（EEG）情绪识别。BiDANN 模型受神经科学发现的启发，即人类大脑的左右半球对情绪反应存在不对称性。该模型包含一个全局域判别器和两个局部域判别器，它们与分类器对抗工作，以学习每个半球的判别性情绪特征。同时，它试图减少源域和目标域之间每个半球的可能域差异，以提高识别模型的泛化能力。此外，我们还提出了一种改进版本的 BiDANN，记为 BiDANN-S，通过降低被试个人信息对 EEG 情绪识别的影响，用于被试无关的 EEG 情绪识别问题。在 SEED 数据库上进行了大量实验，以评估 BiDANN 和 BiDANN-S 的性能。实验结果表明，所提出的 BiDANN 和 BiDANN 模型在 EEG 情绪识别方面达到了最先进的性能。

索引词—脑电图情绪识别，长短期记忆（LSTM），大脑半球不对称性，对抗网络

1 引言

情绪，作为一种常见的心理现象，与人类的认知和行为密切相关[1]。尽管人类可以轻易捕捉情绪，但机器仍难以理解。作为情感计算领域最活跃的研究课题之一[2]，情绪识别已引起计算机视觉和模式识别研究群体的广泛关注。基本上，情绪的响应可分为外部响应和内部响应。典型的外部响应包括人的面部表情、手势或言语，典型的内部响应包括皮肤电导反应、心率、血压、呼吸频率、脑电图（EEG）、脑磁图（MEG）[3]。从神经科学的角度来看[4]，

有一些主要的脑皮层区域，例如眶额叶皮层、腹内侧前额叶皮层和杏仁核，与情绪密切相关[5]，[6]，[7]，这为我们通过记录人脑在这些脑区中的信号来解码情绪提供了一种潜在的方法。例如，通过将脑电图电极放置在头皮上，我们可以记录大脑的神经活动，这些活动可以用于识别人的情绪。

传统的脑电图情绪识别系统通常由两个主要部分组成，即特征提取部分和分类器设计部分。脑电图特征可以从时域、频域或时频域中提取[8]。Jenke 等人[8]对脑电图特征提取方法进行了全面的综述。在处理分类问题时，过去几年中提出了许多脑电图情绪识别模型和方法[9][10]。Zheng 等人[11]提出了一种新的组稀疏典型相关分析（GSCCA）方法，用于同时进行脑电图通道选择和情绪识别。Li 等人[12]提出了一种图正则化稀疏线性回归（GRSLR）方法来处理脑电图情绪识别问题。最近，使用深度学习方法进行脑电图情绪识别已被广泛采用，并显示出比传统方法更好的性能。在[13]中，Zheng 等人提出使用深度信念网络（DBN）进行脑电图情绪分类。在[14]中，Song 等人使用图来对多通道脑电图特征进行建模，然后基于此进行脑电图情绪分类。

尽管针对脑电图情绪识别问题已经提出了许多算法或模型，但大多数模型都集中在训练和测试数据来自同一领域的场景中，我们通常假设训练和测试数据的特征分布是

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similar. Under this assumption, the label information of the target data is predicted directly based on the classifier learned with the source domain data samples. For cross-domain EEG emotion recognition problems, however, many of the aforementioned EEG emotion recognition methods would fail because of the mismatch problem of the EEG feature distributions between the source domain and the target one. A typical example is the cross-subject EEG emotion recognition problem, in which the training and testing EEG data are from different subjects.

To deal with the challenging cross-subject EEG emotion recognition problem, Pandey et al. [15] proposed a subject independent approach for EEG emotion recognition. Li et al. [16] proposed another method for cross-subject EEG emotion recognition. Recently, Ganin et al. proposed a domain adversarial neural networks (DANN) to deal with the domain adaptation problems [17]. It uses a domain discriminator as well as a feature generator to alleviate the differences of the feature distribution between the source domain and the target domain and at the same time generate domain-invariable data features without any class information from testing data. In our preliminary work of [18], we had proposed a novel neural network model that fits the asymmetry of emotional brain into the DANN structure meanwhile utilizing time information for EEG emotion recognition task.

On the other hand, from the neuroscience point of view, it is notable that the left and right hemispheres of human's brain are not entirely symmetrical [19], [20], [21], [22]. In [23], Dimond et al. first found that there exists different 'emotional vision' between the right and the left hemispheres of the human brain, where the right hemisphere demonstrates to be more powerful in perceiving unpleasant emotion than the left one. In [24], [25], [26], Davidson et al. showed that the EEG signals of the left frontal cortex are closely related to positive emotions while the ones in the right frontal cortex are related to negative emotions. In [27], Costanzo et al. focused on the research of mood induction and language activation based on functional magnetic resonance imaging (fMRI) and found that sadness and happiness are processed by each hemisphere. Moreover, some researchers had also proposed methods to measure the differences between the two hemispheres, which were used to the depression detection and EEG monitoring [28], [29], [30]. Nevertheless, it is still an interesting topic of how to utilize the asymmetry property of brain to improve the emotion recognition performance.

Motivated by the recent success of the DANN method and the findings of neuroscience about the asymmetry property of human brain, in this paper we propose a novel deep neural network model, called bi-hemispheres domain adversarial neural network (BiDANN) model, to deal with EEG emotion recognition problem, which can be seen as an extension of our preliminary work of [18]. In the BiDANN model, we design a global and two local domain discriminators that work adversarially with a classifier to learn discriminative emotional features of each hemisphere and at the same time to alleviate the domain differences between source and target domains on each hemisphere. Specifically, the BiDANN model consists of three major modules, i.e., the feature extractor module, the classifier module, and the domain discriminator module. The major goal of feature extractor module is to extract discriminative deep features based on the original EEG features

extracted from each cerebral hemisphere, the classifier module aims to predict the emotion class information based on the EEG deep features, and the target of the domain discriminator module is to alleviate the domain differences between the source and target domains. Moreover, considering that the subject's information, e.g., the personal identity information, may influence the EEG emotion recognition performance, we also propose an improved version of BiDANN, denoted by BiDANN-S, to deal with subject-independent EEG emotion recognition. The basic idea is to ensure that the extracted EEG features are robust to the variation of subjects so as to improve the generality of the recognition model.

The remainder of this paper is organized as follows: In Section 2, we give a brief overview of preliminary work. In Sections 3 and 4, we propose the BiDANN model and the BiDANN-S model, respectively, for EEG emotion recognition. The experiments will be presented in Section 5, and finally, in Section 6, we will conclude the paper.

2 PRELIMINARIES

In this section, we will briefly review the preliminary work of long short term memory (LSTM) method [31] and the domain adversarial neural networks (DANN) method [32].

2.1 LSTM

In the traditional neural networks methods, the inputs are independent with each other and hence they neglect sequence information. In contrast to the traditional neural network methods, the recurrent neural networks (RNN) method [33] aims to map the input sequence into a series of hidden states through a complex dynamics transform with a chain of repeating modules of neural network and remember and forget the information in the whole sequence adaptively [34]. The LSTM network [31] is a special type of RNN, which replaces the repeating module of RNN with a more complex structure consisting of three gates and a cell state to deal with long-term dependence problem.

2.2 DANN

The DANN model was proposed by Ganin et al. [17] to deal with domain adaptation problem under the assumption that the desired features should not be able to discriminate the data samples between the source and target domains. To this end, DANN borrows the idea of generative adversarial networks (GAN) [32] whose goal is to design an adversarial process that simultaneously learns a generative model and a discriminative model, in which the discriminative model estimates the probability of a sample coming from the real data set or a fake generative data set whereas the generative model aims to estimate the distribution of real training data. Based on the GAN method, the DANN method aims to generate domain-invariant data features that are discriminative for the classification task whereas in-discriminative for the shift between the source and target domains, in which a gradient reversal layer (GRL) is introduced for this purpose.

3 BiDANN FOR EEG EMOTION RECOGNITION

In this section, we will first address the BiDANN model and then extend it to deal with subject-independent EEG emotion

相似。在此假设下，直接基于使用源域数据样本学习到的分类器预测目标数据的标签信息。然而，对于跨域脑电图情绪识别问题，由于源域和目标域之间脑电图特征分布的不匹配问题，上述许多脑电图情绪识别方法会失败。一个典型的例子是跨被试脑电图情绪识别问题，其中训练和测试脑电图数据来自不同的被试。

为解决具有挑战性的跨被试脑电图情绪识别问题，Pandey 等人[15]提出了一种被试无关的脑电图情绪识别方法。Li 等人[16]提出了另一种跨被试脑电图情绪识别方法。最近，Ganin 等人提出了一种领域对抗神经网络（DANN）来处理领域自适应问题[17]。它使用领域判别器和特征生成器来缓解源领域和目标领域之间特征分布的差异，同时在不使用测试数据的任何类别信息的情况下生成领域不变的数据特征。在我们的初步工作[18]中，我们提出了一种新型神经网络模型，将情绪大脑的不对称性融入 DANN 结构，同时利用时间信息进行脑电图情绪识别任务。

另一方面，从神经科学角度来看，值得注意的是人类大脑的左右半球并不完全对称[19], [20], [21], [22]。在[23]中，Dimond 等人首次发现人类大脑的左右半球存在不同的“情绪视觉”，其中右半球在感知负面情绪方面比左半球更强大。在[24], [25], [26]中，Davidson 等人表明左额叶皮层的脑电（EEG）信号与积极情绪密切相关，而右额叶皮层的信号则与消极情绪相关。在[27]中，Costanzo 等人基于功能性磁共振成像（fMRI）对情绪诱导和语言激活进行了研究，发现悲伤和快乐分别由每个半球处理。此外，一些研究者还提出了测量两半球差异的方法，这些方法被用于抑郁症检测和脑电监测[28], [29], [30]。尽管如此，如何利用大脑的这种不对称特性来提高情绪识别性能仍然是一个有趣的研究课题。

受 DANN 方法近期成功及神经科学关于人脑不对称性发现的启发，本文提出了一种新型深度神经网络模型，称为双半球域对抗神经网络（BiDANN）模型，用于处理脑电图情绪识别问题，这可视为我们初步工作[18]的扩展。在 BiDANN 模型中，我们设计了一个全局域判别器和两个局部域判别器，它们与分类器对抗工作，以学习每侧大脑半球的判别性情绪特征，同时缓解源域和目标域在每侧大脑半球上的域差异。具体而言，BiDANN 模型由三个主要模块组成，即特征提取模块、分类器模块和域判别器模块。

特征提取模块的主要目标是基于从每个大脑半球提取的原始脑电图特征，提取具有判别性的深度特征；分类器模块旨在基于脑电图深度特征预测情绪类别信息；域判别器模块的目标是减轻源域和目标域之间的域差异。此外，考虑到受试者信息（如个人信息）可能会影响脑电图情绪识别性能，我们还提出了 BiDANN 的改进版本，记为 BiDANN-S，用于处理受试者无关的脑电图情绪识别。基本思想是确保提取的脑电图特征对受试者变化具有鲁棒性，以提高识别模型的泛化性。

本文的其余部分安排如下：在第 2 节，我们简要概述了前期工作。在第 3 节和第 4 节，我们分别提出了用于脑电图情绪识别的 BiDANN 模型和 BiDANN-S 模型。实验将在第 5 节中呈现，最后，在第 6 节，我们将总结全文。

2 前期工作

在本节中，我们将简要回顾长短期记忆（LSTM）方法[31]和领域对抗神经网络（DANN）方法[32]的初步工作。

2.1 LSTM

在传统的神经网络方法中，输入是相互独立的，因此它们忽略了序列信息。与传统神经网络方法不同，循环神经网络（RNN）方法[33]旨在通过一个由重复的神经网络模块链组成的复杂动态变换，将输入序列映射到一系列隐藏状态，并自适应地记忆和遗忘整个序列中的信息[34]。LSTM 网络[31]是一种特殊的 RNN，它用包含三个门和一个单元状态的更复杂结构来替换 RNN 的重复模块，以处理长期依赖问题。

2.2 DANN

DANN 模型由 Ganin 等人[17]提出，用于解决域适应问题，其前提是期望特征不应能区分源域和目标域中的数据样本。为此，DANN 借鉴了生成对抗网络（GAN）[32]的思想，其目标是设计一个对抗过程，同时学习生成模型和判别模型，其中判别模型估计样本来自真实数据集或伪造生成数据集的概率，而生成模型旨在估计真实训练数据的分布。基于 GAN 方法，DANN 方法旨在生成对分类任务具有判别性的域不变数据特征，而对源域和目标域之间的转换不具有判别性，为此引入了梯度反转层（GRL）。

3 双半球域对抗神经网络模型用于脑电图情绪识别

在本节中，我们将首先介绍 BiDANN 模型，然后将其扩展以处理与受试者无关的脑电图情绪

TABLE 1
The Major Notations and the Corresponding Definitions Used in BiDANN

Notation	Definition
$\mathbf{X}$	EEG raw feature matrix
$\mathbf{Y}$	LSTM feature matrix
$\mathbf{H}$	Deep feature matrix
θ	Learnable parameters
$E_f(\cdot)$	Feature extractor function
$L(\cdot)$	Loss function
$\mathcal{D}$	Domain label set

recognition problem. Before addressing the detailed information of the model, we will first provide a list of the major notations and the corresponding definitions in Table 1. In addition, we use the subscript notations of f, c, d to denote the feature extractor, the classifier and the discriminator, respectively. Similarly, we use the superscript notations of l, r to denote left and right hemisphere, respectively, and use the subscript notations of S, T to denote subscript of source and target domain respectively. For example, we use $\mathbf{X}^l$ and $\mathbf{X}_S$ to denote the EEG feature matrix associated with the left hemisphere and the EEG feature matrix associated with the source domain, respectively, and use $\mathbf{X}_S^l$ to denote the EEG feature matrix corresponding to the left hemisphere of the source domain.

3.1 BiDANN

The target of BiDANN is to alleviate the possible feature distribution difference between source and target domains, either in each hemisphere or the whole brain cortex area. To achieve this goal, we borrow the basic idea of the DANN method by leverage the adversarial operation between the source and the target domains into the discriminative EEG feature learning module, and also make use of the neuroscience findings that the left and right hemispheres of human's brain are asymmetric to the emotional response to further enhance the discriminative ability of the EEG features. Fig. 1 illustrates the framework of the BiDANN model, which consists of the following major parts: (1) two feature extractors (f_l and f_r); (2) a global discriminator (D_g) and two local discriminators (D_l and D_r); (3) three GRL, and a classifier (C). In Fig. 1, the black lines and arrows refer to source domain path that the EEG data matrix $\mathbf{X}_S$ is from the source domain, while the red lines and arrows refer to target domain path that the EEG data matrix $\mathbf{X}_T$ is from the target domain.

In BiDANN model, the two feature extractors f_l and f_r aim to capture the dynamic features of two hemispheric EEG signal separately. The three discriminators are trained on a binary domain label set $\mathcal{D} = \{0, 1\}$, in which the domain labels of source samples are set to 0 while the domain labels of target samples are set to 1. The global discriminator aims to narrow the features' distribution gap between source and target domains, while the two local discriminators complementally eliminate the left and right hemispheric features' distribution difference between source and target domains. The use of GRL is to maximize the loss of discriminators by leaving the input data unchanged during forward propagation and reversing the gradient by multiplying it with a negative scalar during backpropagation [17]. The classifier maps

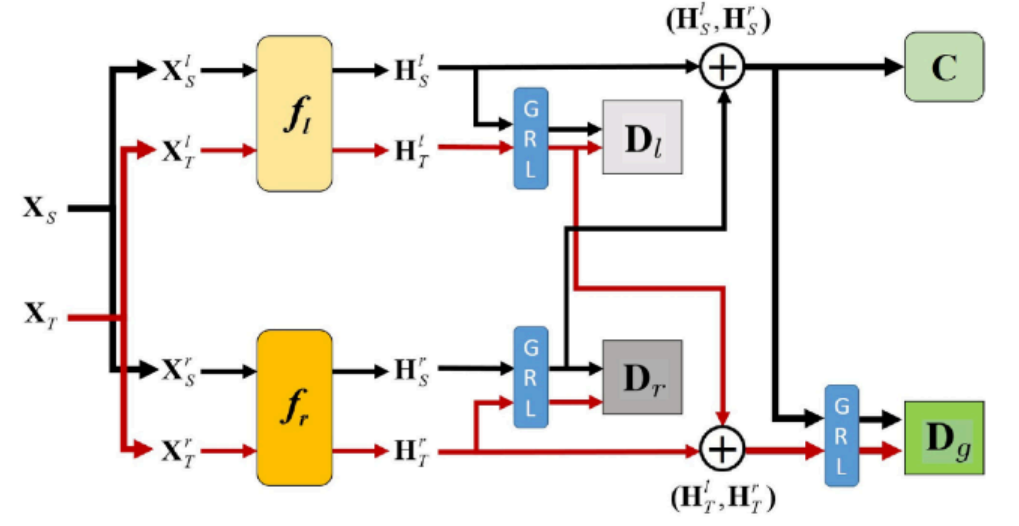


Fig. 1. The framework of BiDANN. The black lines and arrows refer to source domain path, while the red lines and arrows refer to target domain path.

the emotion-related and domain-invariant features into class label space to predict the class labels. In what follows, we will specify the BiDANN model with respect to feature extractor, local discriminator, global discriminator and classifier:

3.1.1 Feature Extractor

The goal of feature extractor is to extract more discriminative features from handcraft EEG features so as to improve the EEG classification performance. The whole feature extraction procedure is summarized in Fig. 2, which consists of the handcraft EEG feature extraction part and the discriminative deep feature learning part.

To extract the handcraft EEG features, we first decompose the EEG signals into five frequency bands, i.e., δ band (1-3 Hz), θ band (4-7 Hz), α band (8-13 Hz), β band (14-30 Hz), and γ band (31-50Hz). For each EEG frequency band, we segment the EEG signals of each trial into some clips using a non-overlapped Hanning window [13], in which the EEG signals of each clip is used to extract 62 differential entropy (DE) features, resulting in a 310-dimensional feature vector. To explore the temporal dependence among the EEG feature vectors, every 9 neighboring EEG feature vectors determined by a slicing window with the size of 9 EEG clips are chosen to form an EEG feature matrix, denoted by $\mathbf{X} \in \mathbb{R}^{d_x \times t_x}$, and this matrix is used to serve as one sample. By slicing the window with an interval of one EEG clip, we finally obtain a set of EEG feature matrices as the input EEG data.

Suppose that $\mathbf{X}^l = [\mathbf{x}_1^l, \dots, \mathbf{x}_{t_x}^l] \in \mathbb{R}^{d_x^l \times t_x}$ and $\mathbf{X}^r = [\mathbf{x}_1^r, \dots, \mathbf{x}_{t_x}^r] \in \mathbb{R}^{d_x^r \times t_x}$ denote two handcraft EEG feature matrices of an EEG clip sequence associated with the left and right hemisphere, respectively, where d_x^l and d_x^r denote the dimensions of the EEG feature vector, and t_x is the number of EEG clips. Let

$$\mathbf{X} = [\mathbf{X}^l, \mathbf{X}^r] = [\mathbf{x}_1^l, \dots, \mathbf{x}_{t_x}^l, \mathbf{x}_1^r, \dots, \mathbf{x}_{t_x}^r].$$

To capture the temporal dependence among the EEG feature sequence, we adopt the LSTM model to extract more discriminative EEG features, resulting in the following LSTM feature matrices $\mathbf{Y}^l$ and $\mathbf{Y}^r$:

$$\mathbf{Y}^l = [\mathbf{y}_1^l, \dots, \mathbf{y}_{t_x}^l] = [f(\mathbf{x}_1^l), \dots, f(\mathbf{x}_{t_x}^l)], \quad (1)$$

$$\mathbf{Y}^r = [\mathbf{y}_1^r, \dots, \mathbf{y}_{t_x}^r] = [f(\mathbf{x}_1^r), \dots, f(\mathbf{x}_{t_x}^r)], \quad (2)$$

表 1 主要符号及其对应

BiDANN 中使用的定义	
符号	定义
X	脑电图原始特征矩阵
Y	LSTM 特征矩阵
H	深度特征矩阵
u	可学习参数
$E \partial p$	特征提取函数
$L \partial p$	损失函数
D	领域标签集

识别问题。在介绍模型的详细信息之前，我们将首先在表 1 中列出主要符号及其定义。此外，我们使用 f、c、d 的下标符号分别表示特征提取器、分类器和判别器。类似地，我们使用 l、r 的上标符号分别表示左半球和右半球，并使用 S、T 的下标符号分别表示源域和目标域的下标。例如，我们使用 X^l 和 X^r 分别表示与左半球相关的脑电图特征矩阵和与源域相关的脑电图特征矩阵，并使用 X^l 表示与源域左半球对应的脑电图特征矩阵。

3.1 BiDANN

BiDANN 的目标是缓解源域和目标域之间可能的特征分布差异，无论是在每个半球还是整个大脑皮层区域。为了实现这一目标，我们借鉴了 DANN 方法的基本思想，通过将源域和目标域之间的对抗操作引入到判别性 EEG 特征学习模块中，并利用神经科学发现——人类大脑的左右半球对情绪反应存在不对称性——来进一步增强 EEG 特征的判别能力。图 1 展示了 BiDANN 模型的框架，该模型由以下主要部分组成：(1)两个特征提取器 (f_l 和 f_r)；(2)一个全局判别器 (D) 和两个局部判别器 (D_l 和 D_r)；(3)三个 GRL，以及一个分类器 (C)。在图 1 中，黑线和箭头表示 EEG 数据矩阵 X 来自源域的源域路径，而红线和箭头表示 EEG 数据矩阵 X 来自目标域的目标域路径。

在 BiDANN 模型中，两个特征提取器 f_l 和 f_r 分别旨在捕获两个半球脑电图信号的动态特征。三个判别器在二元域标签集 $D=\{0,1\}$ 上训练，其中源样本的域标签设置为 0，而目标样本的域标签设置为 1。全局判别器旨在缩小源域和目标域之间特征分布的差距，而两个局部判别器互补地消除源域和目标域之间左右半球特征分布的差异。使用 GRL 的目的是通过在正向传播过程中保持输入数据不变，并在反向传播过程中通过乘以一个负标量来反转梯度，从而最大化判别器的损失[17]

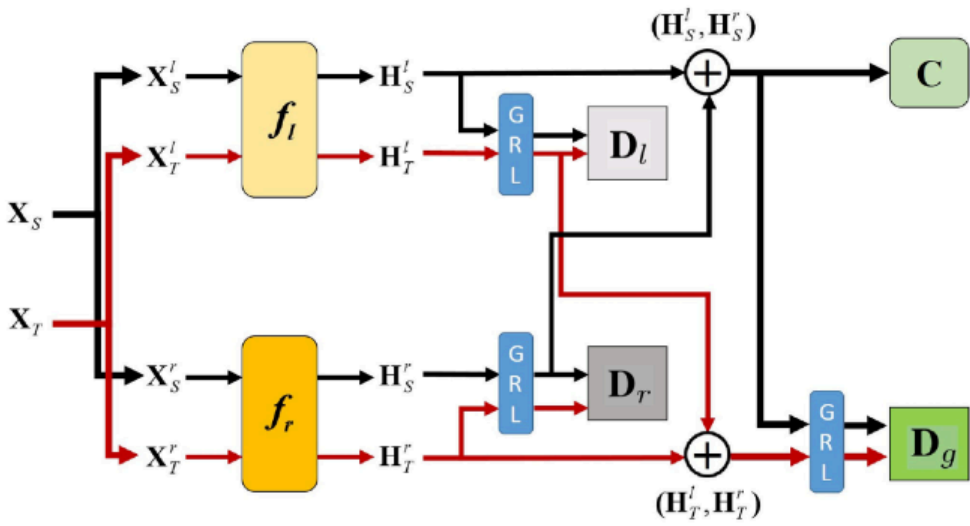


图 1. BiDANN 框架。黑线和箭头表示源域路径，红线和箭头表示目标域路径。

分类器将情绪相关和域不变特征映射到类别标签空间以预测类别标签。在下文中，我们将针对特征提取器、局部判别器、全局判别器和分类器具体说明 BiDANN 模型：

3.1.1 特征提取器

特征提取器的目标是从手工制作的脑电图特征中提取更具判别性的特征，以提高脑电图分类性能。整个特征提取过程如图 2 所示，它由手工制作的脑电图特征提取部分和判别性深度特征学习部分组成。为了提取手工制作的脑电图特征，我们首先将脑电图信号分解为五个频段，即 d 频段 (1-3 Hz)、u 频段 (4-7 Hz)、a 频段 (8-13 Hz)、b 频段 (14-30 Hz) 和 g 频段 (31-50Hz)。对于每个脑电图频段，我们使用非重叠的汉宁窗[13]将每个试验的脑电图信号分割成一些片段，其中每个片段的脑电图信号用于提取 62 个差分熵 (DE) 特征，从而得到一个 310 维的特征向量。为了探索脑电图特征向量之间的时序依赖性，我们选择由一个包含 9 个脑电图片段大小的切片窗确定的 9 个相邻脑电图特征向量，形成一个脑电图特征矩阵，记为 $X \in \mathbb{R}^{t \times d}$ ，并将该矩阵用作一个样本。通过以一个脑电图片段为间隔移动窗口，我们最终得到一组脑电图特征矩阵作为输入脑电图数据。假设 $X = (x_1, \dots, x_n) \in \mathbb{R}^{t \times d}$ 和 $X = (x_1, \dots, x_n) \in \mathbb{R}^{t \times d}$ 分别表示与左右半球相关的脑电图片段序列的两个手工制作脑电图特征矩阵，其中 d 和 d 表示脑电图特征向量的维度，t 是脑电图片段的数量。令

$$X = \frac{1}{2}X; X\check{S} = \frac{1}{2}x; \dots; x; x; \dots; x\check{S};$$

为了捕捉脑电图特征序列中的时间依赖性，我们采用 LSTM 模型来提取更具判别性的脑电图特征，从而得到以下 LSTM 特征矩阵 Y 和 Y：

$$Y = \frac{1}{2}y; \dots; y\check{S} = \frac{1}{2}f(x); \dots; f(x)\check{S}; \tag{1}$$

$$Y = \frac{1}{2}y; \dots; y\check{S} = \frac{1}{2}f(x); \dots; f(x)\check{S}; \tag{2}$$

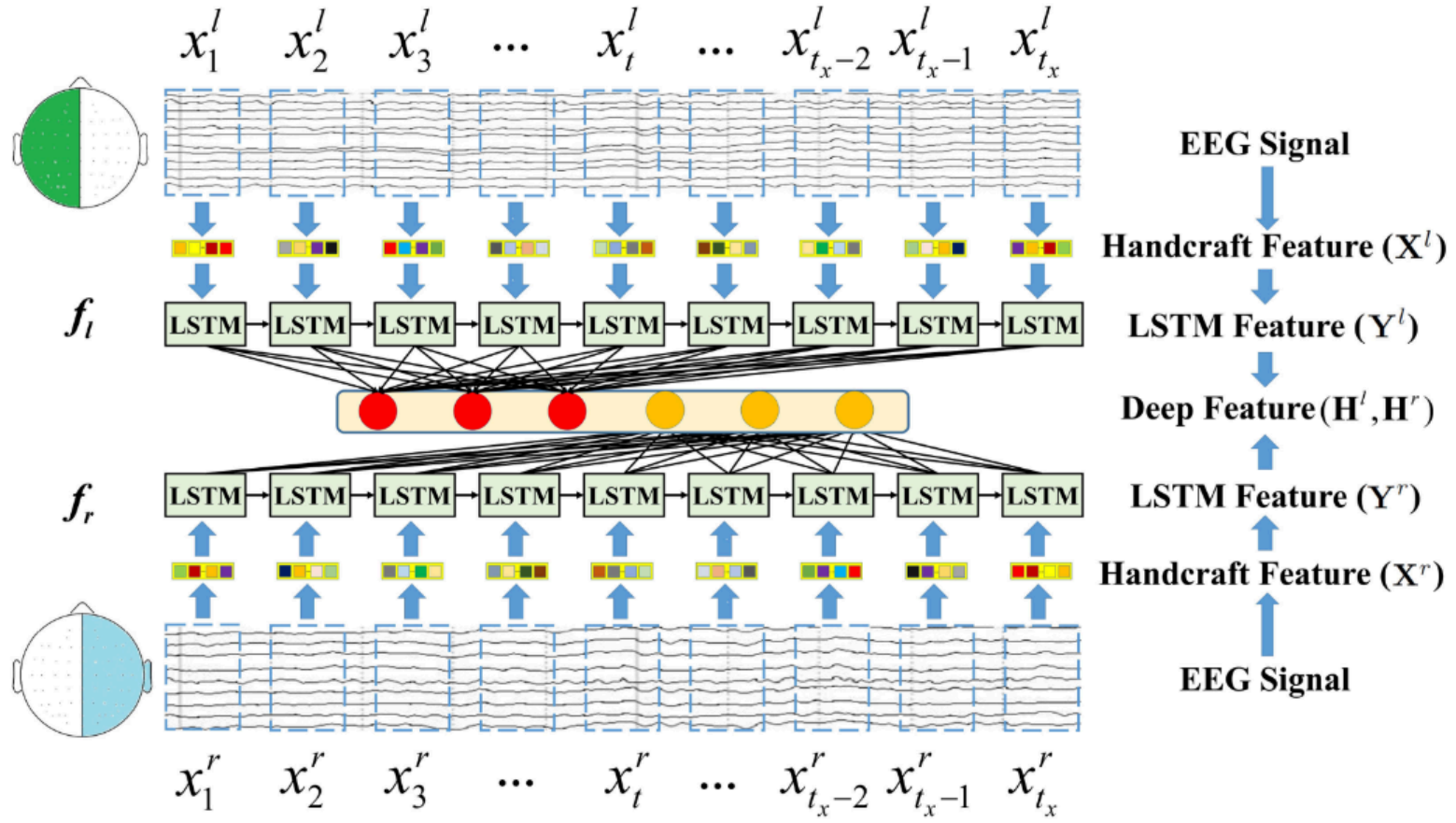


Fig. 2. The feature extraction procedures in the BiDANN model, which consists of the handcraft EEG feature extraction part and the discriminative deep feature extraction part.

where $f(\cdot)$ denote the LSTM feature learning operation, $\mathbf{y}_i^l = f(\mathbf{x}_i^l)$ and $\mathbf{y}_i^r = f(\mathbf{x}_i^r)$ ($i = 1, \dots, t_x$).

Let $\mathbf{G} = [G_{ik}]_{t_x \times K}$ be a $t_x \times K$ coefficient matrix and $\sigma(\cdot)$ be a nonlinear mapping. Then, under the transformation of $\mathbf{G}$ and the operation of $\sigma(\cdot)$, we obtain the following deep feature matrices $\mathbf{H}^l$ and $\mathbf{H}^r$:

$$\mathbf{H}^l = \sigma(\mathbf{Y}^l \mathbf{G} + \mathbf{B}) = \sigma(f(\mathbf{X}^l) \mathbf{G} + \mathbf{B}), \quad (3)$$

$$\mathbf{H}^r = \sigma(\mathbf{Y}^r \mathbf{G} + \mathbf{B}) = \sigma(f(\mathbf{X}^r) \mathbf{G} + \mathbf{B}), \quad (4)$$

where $\mathbf{B}$ is a bias matrix.

Now suppose that $\mathbf{X}_S^l$ ($\mathbf{X}_S^r$) and $\mathbf{X}_T^l$ ($\mathbf{X}_T^r$) are the input EEG feature matrices of the left (right) hemisphere associated with the source domain and the target domain, respectively. Let $\mathbf{H}_S^l$ ($\mathbf{H}_S^r$) and $\mathbf{H}_T^l$ ($\mathbf{H}_T^r$) be the corresponding deep feature matrices obtained using (3) and (4). Then, by combining $\mathbf{H}_S^l$ ($\mathbf{H}_S^r$) with $\mathbf{H}_T^l$ ($\mathbf{H}_T^r$), we obtain the following new matrices:

$$\mathbf{H}_{S,T}^l = [\mathbf{H}_S^l, \mathbf{H}_T^l] = E_f(\mathbf{X}_S^l, \mathbf{X}_T^l), \quad (5)$$

$$\mathbf{H}_{S,T}^r = [\mathbf{H}_S^r, \mathbf{H}_T^r] = E_f(\mathbf{X}_S^r, \mathbf{X}_T^r), \quad (6)$$

where $E_f(\cdot)$ denotes the feature extractor function.

3.1.2 Local, Global Discriminators and Classifier

The use of local discriminator aims to alleviate the feature distribution difference between source and target domains, in either left hemisphere or right one. In contrast to the local discriminators, the target of global discriminator is also to eliminate the feature distribution difference between source and target domains, but across the whole brain cortex area instead of only the hemisphere area.

Different from both local and global discriminators, the purpose of the classifier part is to learn the discriminative features across source and target domains, which is implemented by applying the softmax function to the transformed hidden states to predict the class label, where the loss function of class label prediction is expressed as:

$$L_c(\mathbf{X}_S; \theta_f^l, \theta_f^r, \theta_c) = \sum_c -\tau(y, c) \times \log P(c|\mathbf{X}_S), \quad (7)$$

where c is the label of c -th emotion class, y is the ground-truth label associated with the data sample $\mathbf{X}_S$, and

$$\tau(y, c) = \begin{cases} 1, & \text{if } y = c, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

The optimal parameters of feature extractors are determined by minimizing the loss of classifier and maximizing the loss of both local and global domain discriminators. This leads to an adversarial learning between classifier and domain discriminators for emotion-related and domain-invariant EEG features.

According to the above analysis, we obtain that the BiDANN problem can be formulated as the optimization problem of minimizing the overall loss function $L(\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g)$ defined as the following form:

$$\begin{aligned} \min L(\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g) \\ = L_c(\mathbf{X}_S; \theta_f^l, \theta_f^r, \theta_c) - L_d^l(\mathbf{X}_{S,T}^l; \theta_f^l, \theta_d^l) \\ - L_d^r(\mathbf{X}_{S,T}^r; \theta_f^r, \theta_d^r) - L_d^g(\mathbf{X}_{S,T}; \theta_f^l, \theta_f^r, \theta_d^g), \end{aligned} \quad (9)$$

where $L_c(\cdot)$, $L_d^l(\cdot)$, $L_d^r(\cdot)$ and $L_d^g(\cdot)$ denote the loss function of classifier, the loss function of local discriminator associated with the left and right hemisphere, and the loss function of global discriminator, respectively, and the parameters θ_c , θ_d^l , θ_d^r , θ_d^g , θ_f^l and θ_f^r denote the corresponding model parameters.

3.2 Optimization of BiDANN

We can iteratively train the classifier and three discriminators and update the parameters with the similar approach of standard deep learning methods by chain rule. Specifically, to solve the optimal solution of BiDANN, we adopt the stochastic gradient descent (SGD) algorithm [35] to find optimal model parameters $(\theta_f^l, \theta_f^r, \theta_c)$ in (9). Moreover, we also

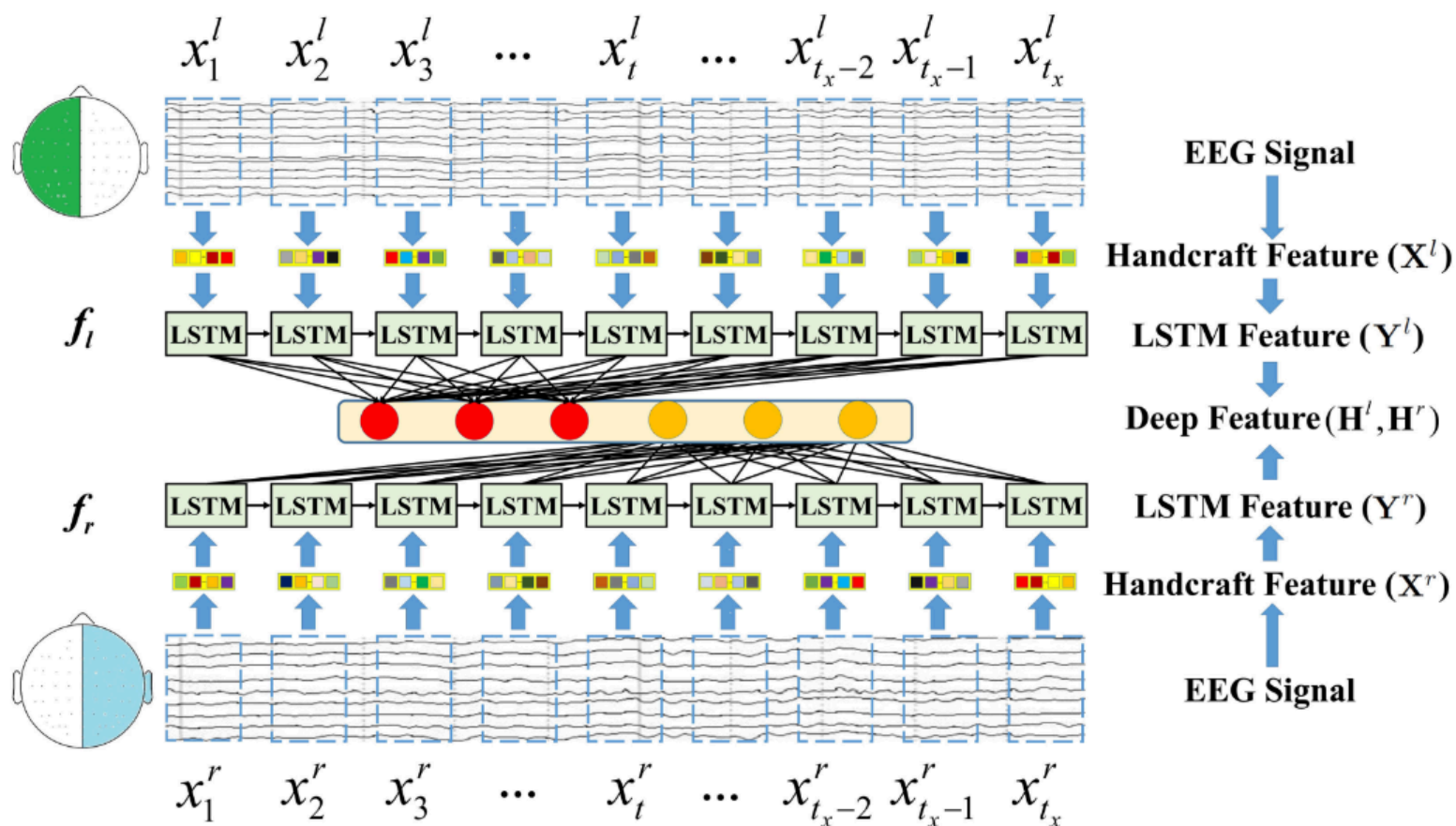


图 2. BiDANN 模型中的特征提取过程，该模型由手工 EEG 特征提取部分和判别深度特征提取部分组成。

其中 f_{∂} 表示 LSTM 特征学习操作， $y_{\partial} f_{\partial} x_{\partial}$ 和 $y_{\partial} f_{\partial} x_{\partial}$ ($i = 1; \dots$;

t)。设 $G = \frac{1}{2}G_{\partial}$ 为一个 $t \times K$ 系数矩阵， s_{∂} 为一个非线性映射。在 G 的变换和 s_{∂} 的操作下，我们得到以下深度特征矩阵 Hand H:

$$H = \frac{1}{2} s_{\partial} Y G_{\partial} B_{\partial} = \frac{1}{2} s_{\partial} f_{\partial} X_{\partial} G_{\partial} B_{\partial}; \quad (3)$$

$$H = \frac{1}{2} s_{\partial} Y G_{\partial} B_{\partial} = \frac{1}{2} s_{\partial} f_{\partial} X_{\partial} G_{\partial} B_{\partial}; \quad (4)$$

其中 B 是一个偏差矩阵。

现在假设 $X(X)$ 和 $X(X)$ 分别是左侧 (右侧) 半球的源域和目标域的输入脑电图特征矩阵。令 $H(H)$ 和 $H(H)$ 是使用公式 (3) 和 (4) 获得的相关深度特征矩阵。然后，通过结合 $H(H)$ 和 $H(H)$ ，我们得到以下新矩阵:

$$H = \frac{1}{2} H; H_{\partial} = \frac{1}{2} E_{\partial} X; X_{\partial}; \quad (5)$$

$$H = \frac{1}{2} H; H_{\partial} = \frac{1}{2} E_{\partial} X; X_{\partial}; \quad (6)$$

其中 E_{∂} 表示特征提取函数。

3.1.2 局部判别器、全局判别器和分类器

使用局部判别器的目的是缓解源域和目标域在左半球或右半球中的特征分布差异。与局部判别器不同，全局判别器的目标也是消除源域和目标域之间的特征分布差异，但范围是整个大脑皮层区域，而不仅仅是半球区域。

与局部判别器和全局判别器不同，分类器的目的是学习跨源域和目标域的判别特征，这是通过将转换后的隐藏状态应用于 softmax 函数来预测类别标签实现的，其中类别标签预测的损失函数表示为:

$$L_{\partial} X; u; u; u_{\partial} = \sum_c X_{\partial} t_{\partial} y; c_{\partial} \log P_{\partial} c_{\partial} X_{\partial}; \quad (7)$$

其中 c 是第 c 类情绪类别的标签， y 是与数据样本 X 相关的真实标签，并且

$$t_{\partial} y; c_{\partial} = \begin{cases} 1; & \text{如果 } y = c; \\ 0; & \text{否则;} \end{cases} \quad (8)$$

特征提取器的最优参数是通过最小化分类器的损失并最大化局部和全局域判别器的损失来确定的。这导致了分类器和域判别器之间针对与情绪相关的和域不变的脑电图特征进行对抗学习。

根据上述分析，我们得出 BiDANN 问题可以表述为最小化整体损失函数 $L(u; u; u; u; u; u)$ 的优化问题，该损失函数定义如下形式:

$$\min L_{\partial} u; u; u; u; u; u = \frac{1}{4} L_{\partial} X; u; u; u_{\partial} L_{\partial} X; u; u_{\partial} L_{\partial} X; u; u_{\partial} L_{\partial} X; u; u; u_{\partial}$$

其中 L_{∂} 、 L_{∂} 、 L_{∂} 和 L_{∂} 分别表示分类器的损失函数、与左右半球相关的局部判别器的损失函数以及全局判别器的损失函数，参数 u 、 u 、 u 、 u 、 u 和 u 表示相应的模型参数。

3.2 BiDANN 优化

我们可以通过链式法则以标准深度学习类似的方式迭代训练分类器和三个判别器并更新参数。具体来说，为了解决 BiDANN 的最优解，我们采用随机梯度下降 (SGD) 算法[35]来寻找(9)中的最优模型参数 (u, u, u) 。此外，我们还

introduce three gradient reversal layers for the three domain discriminators (see Fig. 1), which keep the gradient sign but reverse it while performing the back-propagation operation.

By iteratively minimizing the loss function of L defined in (9), we can finally obtain the optimal parameters. More specifically, suppose that the optimal parameters of $\hat{\theta}_d^g$, $\hat{\theta}_d^l$ and $\hat{\theta}_d^r$ are obtained. Then, the optimal parameters of $\hat{\theta}_f^l$, $\hat{\theta}_f^r$, $\hat{\theta}_c$ can be obtained by minimizing the loss function $L(\mathbf{X}_{S,T}; (\theta_f^l, \theta_f^r, \theta_c), \hat{\theta}_d^g, \hat{\theta}_d^l, \hat{\theta}_d^r)$, i.e.,

$$(\hat{\theta}_f^l, \hat{\theta}_f^r, \hat{\theta}_c) = \arg \min_{\theta_f^l, \theta_f^r, \theta_c} L(\mathbf{X}_{S,T}; (\theta_f^l, \theta_f^r, \theta_c), \hat{\theta}_d^g, \hat{\theta}_d^l, \hat{\theta}_d^r).$$

On the other hand, after obtaining the optimal parameters of $\hat{\theta}_f^l$, $\hat{\theta}_f^r$, $\hat{\theta}_c$, the optimal parameters of $\hat{\theta}_d^g$, $\hat{\theta}_d^l$, and $\hat{\theta}_d^r$ can be updated via solving the following optimization problems:

$$\hat{\theta}_d^g = \arg \max_{\theta_d^g} L(\mathbf{X}_{S,T}; \hat{\theta}_f^l, \hat{\theta}_f^r, \hat{\theta}_c, \theta_d^g, \hat{\theta}_d^l, \hat{\theta}_d^r), \quad (10)$$

$$\hat{\theta}_d^l = \arg \max_{\theta_d^l} L(\mathbf{X}_{S,T}^l; \hat{\theta}_f^l, \hat{\theta}_f^r, \hat{\theta}_c, \theta_d^l, \hat{\theta}_d^g, \hat{\theta}_d^r), \quad (11)$$

$$\hat{\theta}_d^r = \arg \max_{\theta_d^r} L(\mathbf{X}_{S,T}^r; \hat{\theta}_f^l, \hat{\theta}_f^r, \hat{\theta}_c, \theta_d^r, \hat{\theta}_d^g, \hat{\theta}_d^l). \quad (12)$$

Here it should be noted that the parameters before the GRL module (i.e., the parameters of feature extractors) will take the reverse direction of the gradients during back-propagation operation so as to generate data representations that minimize the loss of classifier while maximize the loss of discriminators. The optimization procedure of BiDANN is summarized in Algorithm 1.

Algorithm 1. Procedures of Training BiDANN Model

- Input:** Source data set $\{\mathbf{X}_S\}$ and Target data set $\{\mathbf{X}_T\}$;
Ground-truth label set $\mathbf{L}_S = \{y\}$ of source data set;
Source domain label set $\mathcal{D}_S = [\mathcal{D}_S^l, \mathcal{D}_S^r] = \{0\}$ and target domain label set $\mathcal{D}_T = [\mathcal{D}_T^l, \mathcal{D}_T^r] = \{1\}$;
- 1: Initialize model parameters and learning rate α ;
 - 2: **repeat**
 - 3: Update the parameters of the classifier:
 $\theta_c \leftarrow \theta_c - \alpha \frac{\partial L_c}{\partial \theta_c}$, $\theta_f^l \leftarrow \theta_f^l - \alpha \frac{\partial L_c}{\partial \theta_f^l}$, and $\theta_f^r \leftarrow \theta_f^r - \alpha \frac{\partial L_c}{\partial \theta_f^r}$;
 - 4: Update the parameters of the global discriminator:
 $\theta_d^g \leftarrow \theta_d^g - \alpha \frac{\partial L_d^g}{\partial \theta_d^g}$, $\theta_f^l \leftarrow \theta_f^l + \alpha \frac{\partial L_d^g}{\partial \theta_f^l}$, and $\theta_f^r \leftarrow \theta_f^r + \alpha \frac{\partial L_d^g}{\partial \theta_f^r}$;
 - 5: Update the parameters of the local discriminator corresponding to the left hemisphere:
 $\theta_d^l \leftarrow \theta_d^l - \alpha \frac{\partial L_d^l}{\partial \theta_d^l}$, $\theta_f^l \leftarrow \theta_f^l + \alpha \frac{\partial L_d^l}{\partial \theta_f^l}$;
 - 6: Update the parameters of the local discriminator corresponding to the right hemisphere:
 $\theta_d^r \leftarrow \theta_d^r - \alpha \frac{\partial L_d^r}{\partial \theta_d^r}$, $\theta_f^r \leftarrow \theta_f^r + \alpha \frac{\partial L_d^r}{\partial \theta_f^r}$;
 - 7: If the model has been trained for 100 epochs, then $\alpha \leftarrow 0.9 \times \alpha$ and goto step 3;
 - 8: **until** The iterations satisfies the predefined condition, i.e., the loss function satisfies:

$$L(\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g) < 10^{-3}.$$

Output:

- 9: Parameters: $\hat{\theta}_f^l, \hat{\theta}_f^r, \hat{\theta}_c, \hat{\theta}_d^l, \hat{\theta}_d^r, \hat{\theta}_d^g$.

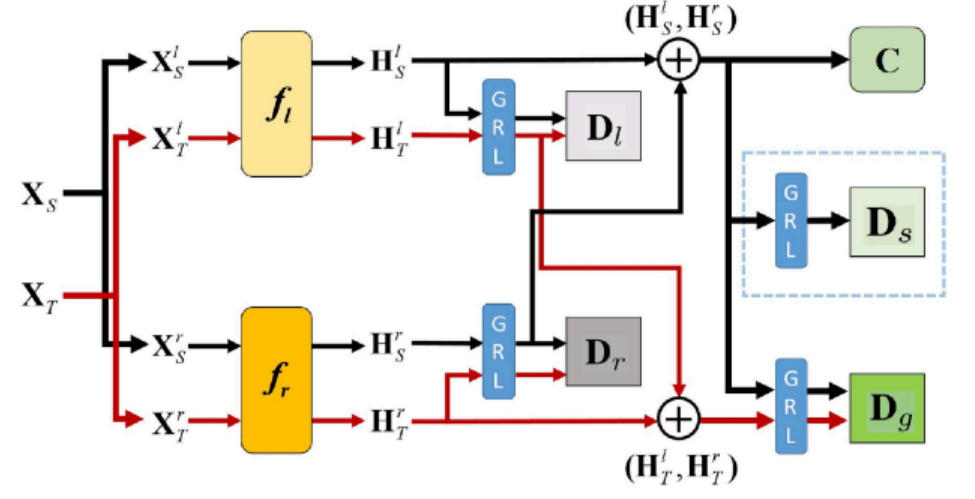


Fig. 3. The framework of the proposed BiDANN-S. Compared with BiDANN, BiDANN-S has an additional subject discriminator to decrease the effect of identity information to the discriminative emotion features.

4 BiDANN-S FOR SUBJECT-INDEPENDENT EEG EMOTION RECOGNITION

The aforementioned BiDANN model provides a useful way to deal with the mismatching feature distribution problem between source domain and the target one. Nevertheless, it is notable that we may still suffer from the difficulties of handling the cross-subject EEG emotion recognition problem, in which the EEG feature distributions of different subjects would be different even if these subjects are from the same source domain. Consequently, it is very desirable to design an effective subject-independent model for the cross-subject EEG emotion recognition problem.

To handle such a subject-independent EEG emotion recognition problem, in this section we propose the BiDANN-S method based on the aforementioned BiDANN model. The basic idea of BiDANN-S is to lower the influences of the personal information of subjects to the EEG emotion recognition as more as possible. This goal is achieved by imposing an additional subject discriminator (denoted by D_s) onto the BiDANN model, in which the subject discriminator tries to lower the discrepancy for the recognition of different subjects. In designing the subject discriminator, we use the subject identity of the labeled source EEG data to serve as another class label set, denoted by $\mathcal{D}_s = \{1, 2, \dots, N\}$, where N is the number of subject in training dataset. In this case, the subject discriminator will work adversarially with the emotion classifier in the BiDANN-S model to result in the subject-independent discriminative EEG features. Fig. 3 illustrates the subject-independent framework of BiDANN-S.

As a result, the loss function of the BiDANN-S model contains an additional part, denoted by $L_d^s(\mathbf{X}_{S,T}; \theta_f^l, \theta_f^r, \theta_d^s)$, compared with the BiDANN method. To achieve the subject-independent features, we should maximize the loss function $L_d^s(\mathbf{X}_{S,T}; \theta_f^l, \theta_f^r, \theta_d^s)$, where θ_d^s is the parameter of subject discriminator. In this case, the entire loss function of BiDANN-S can be expressed as following form:

$$\begin{aligned} L(\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g, \theta_d^s) &= L_c(\mathbf{X}_S; \theta_f^l, \theta_f^r, \theta_c) - L_d^l(\mathbf{X}_{S,T}^l; \theta_f^l, \theta_d^l) \\ &\quad - L_d^r(\mathbf{X}_{S,T}^r; \theta_f^r, \theta_d^r) - L_d^g(\mathbf{X}_{S,T}; \theta_f^l, \theta_f^r, \theta_d^g) \\ &\quad - L_d^s(\mathbf{X}_{S,T}; \theta_f^l, \theta_f^r, \theta_d^s). \end{aligned} \quad (13)$$

为三个域判别器引入三个梯度反转层（见图1），这些层在反向传播操作中保持梯度符号但将其反转。

通过迭代最小化（9）中定义的 L 的损失函数，我们最终可以得到最优参数。更具体地说，假设已经得到了 $\hat{u}$ 、 $\hat{u}$ 和 $\hat{u}$ 的最优参数。然后，通过最小化损失函数 $L(\hat{X}; \delta u; u; u_P; \hat{u}; \hat{u}; \hat{u}_P)$ ，即可得到 $\hat{u}$ 、 $\hat{u}$ 、 $\hat{u}$ 的最优参数，即：

$$\hat{u}; \hat{u}; \hat{u}_P \leftarrow \arg \min_{u; u_P; \hat{u}; \hat{u}; \hat{u}_P} L(\hat{X}; \delta u; u; u_P; \hat{u}; \hat{u}; \hat{u}_P)$$

另一方面，在获得 $\hat{u}$ 、 $\hat{u}$ 、 $\hat{u}$ 的最优参数后，可以通过解决以下优化问题来更新 $\hat{u}$ 、 $\hat{u}$ 和 $\hat{u}$ 的最优参数：

$$\hat{u} \leftarrow \arg \max_{u_d} L(\hat{X}; \hat{u}; \hat{u}; \hat{u}; u; \hat{u}; \hat{u}_P); \quad (10)$$

$$\hat{u} \leftarrow \arg \max_{u_d} L(\hat{X}; \hat{u}; \hat{u}; \hat{u}; \hat{u}; u; \hat{u}_P); \quad (11)$$

$$\hat{u} \leftarrow \arg \max_{u_d} L(\hat{X}; \hat{u}; \hat{u}; \hat{u}; \hat{u}; \hat{u}; u_P); \quad (12)$$

在此应注意的是，GRL 模块之前的参数（即特征提取器的参数）在反向传播操作中会取梯度方向的相反方向，以便生成数据表示，从而最小化分类器的损失，同时最大化判别器的损失。BiDANN 的优化过程总结在算法1中。

算法1. 训练 BiDANN 模型的步骤 输入：源数据集 fX_g 和目标数据集 fX_g ；源数据集的真实标签集 $L \leftarrow fY_g$ ；源域标签集 $D \leftarrow \frac{1}{2}D$ ； $D_S \leftarrow \frac{1}{4}f0g$ 和目标域标签集 $D \leftarrow \frac{1}{2}D$ ； $D_S \leftarrow \frac{1}{4}f1g$ ； 1: 初始化模型参数和学习率 a ； 2: 重复 3: 更新分类器的参数：

$u \leftarrow u + a \cdot \frac{\partial L}{\partial u}$ ， $u \leftarrow u + a \cdot \frac{\partial L}{\partial u}$ ，和

$u \leftarrow u + a \cdot \frac{\partial L}{\partial u}$ ；

4: 更新全局判别器的参数：

$u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$ ， $u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$ ，and
 $u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$

5: 更新本地判别器的参数

对应左半球的：

$u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$ ， $u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$ ；

6: 更新本地判别器的参数

对应右侧脑半球：

$u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$ ， $u \leftarrow u + a \cdot \frac{\partial L}{\partial u_d}$

7: 如果模型已经训练了 100 个 epoch，那么

$a \leftarrow 0.9 \cdot a$ 并转到步骤 3； 8: 直到迭代满足预定义的条件，即，

损失函数满足：

$$L(\hat{u}; u; u; u; u; u_P) < 10;$$

输出：

9: 参数： $\hat{u}$ ； $\hat{u}$ ； $\hat{u}$ ； $\hat{u}$ ； $\hat{u}$ ； $\hat{u}$ 。

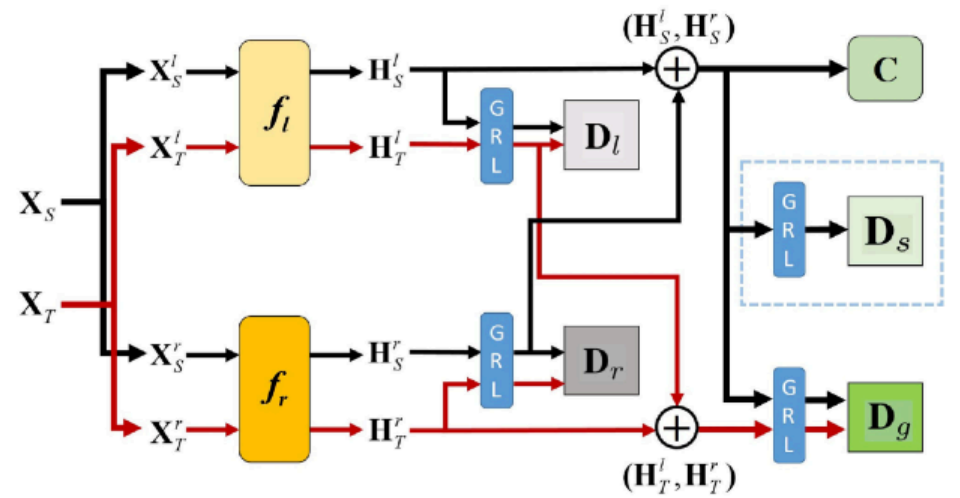


图3. 所提出的 BiDANN-S 框架。与 BiDANN 相比，BiDANN-S 增加了一个受试者判别器，以减少身份信息对判别性情绪特征的影响。

BiDANN-S 用于跨被试 EEG 情绪识别 上述 BiDANN 模型提供了一种处理源域和目标域之间特征分布不匹配问题的有效方法。然而，值得注意的是，我们可能仍然面临处理跨被试 EEG 情绪识别问题的困难，即使这些被试来自同一个源域，不同被试的 EEG 特征分布也会有所不同。因此，设计一个有效的跨被试 EEG 情绪识别的独立模型是非常有必要的。

为了处理这种与被试无关的脑电图情绪识别问题，在本节中我们基于上述 BiDANN 模型提出了 BiDANN-S 方法。BiDANN-S 的基本思想是尽可能降低被试个人信息对脑电图情绪识别的影响。这一目标通过在 BiDANN 模型上增加一个额外的被试判别器（用 D 表示）来实现，该被试判别器试图降低对不同被试识别的差异。在被试判别器的设计中，我们使用标记源脑电图数据的被试身份作为另一个类标签集，记为 $D=\{1, 2, \dots, N\}$ ，其中 N 是训练数据集中被试的数量。在这种情况下，被试判别器将在 BiDANN-S 模型中与情绪分类器对抗，以产生与被试无关的判别性脑电图特征。图3展示了 BiDANN-S 的被试无关框架。

因此，与 BiDANN 方法相比，BiDANN-S 模型的损失函数包含一个额外的部分，记为 $L(X; u; u; u)$ 。为了实现受试者无关的特征，我们应该最大化损失函数 $L(X; u; u; u)$ ，其中 u 是受试者判别器的参数。在这种情况下，BiDANN-S 的整个损失函数可以表示为以下形式：

$$L(u; u; u; u; u; u; u)$$

$$\frac{1}{4} L(X; u; u; u) L(X; u; u) L(X; u; u) L(X; u; u; u) L(X; u; u; u); \quad (13)$$

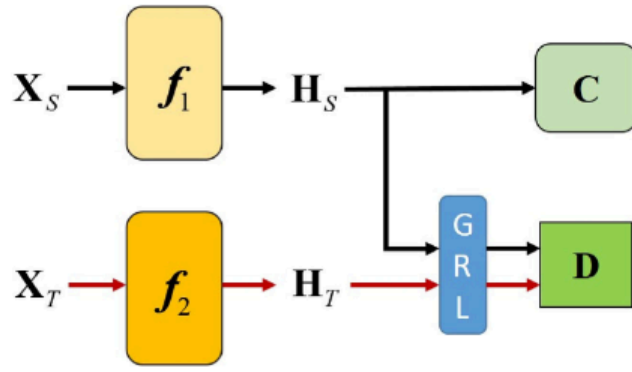


Fig. 4. The framework of BiDANN-R1, which uses two feature extractors to capture the emotion information from source and target data.

By minimizing the loss function $L(\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g, \theta_d^s)$ in (9), we can iteratively solve the optimal model parameters $\theta_f^l, \theta_f^r, \theta_c, \theta_d^l, \theta_d^r, \theta_d^g, \theta_d^s$ of BiDANN-S. The solution approach is similar with that shown in Algorithm 1.

5 EXPERIMENTS

In this section, we will conduct extensive experiments on the SJTU Emotion EEG Dataset (SEED) [13] to evaluate the proposed BiDANN and BiDANN-S methods in EEG emotion recognition. The SEED database contains 15 subjects' EEG signals recorded from 62 electrode channels using ESI NeuroScan with a sampling rate of 1000 Hz. The EEG data are collected when the participants are watching three kinds of emotional film clips, i.e., positive emotion, neutral emotion, negative emotion. To evaluate the proposed BiDANN method, we adopt both subject-dependent and subject-independent strategy to design experiments. In the subject-dependent experiment, we follow the same experimental protocol as that of [13] by choosing the EEG signals associated with two of the three sessions for each subject to serve as the experimental data, in which each session contains 15 trials of EEG signals. In the subject-independent experiment, on the other hand, we also adopt the same experimental protocol as that of [36] by choosing one of the EEG signals associated with one of the three sessions to serve as the experimental data.

Moreover, to evaluate the contributions of different parts of BiDANN to the EEG emotion recognition, we also propose two simplified versions of BiDANN, denoted by BiDANN-R1 and BiDANN-R2, respectively, and compare the recognition performance among them under the same experimental settings. In the BiDANN-R1 model, we use two feature extractors to capture the emotion information from source and target data without considering the hemispheric differences. In contrast to the BiDANN-R1 model, the BiDANN-R2 model uses two feature extractors to capture each brain hemispheric emotion information from source and target data separately.

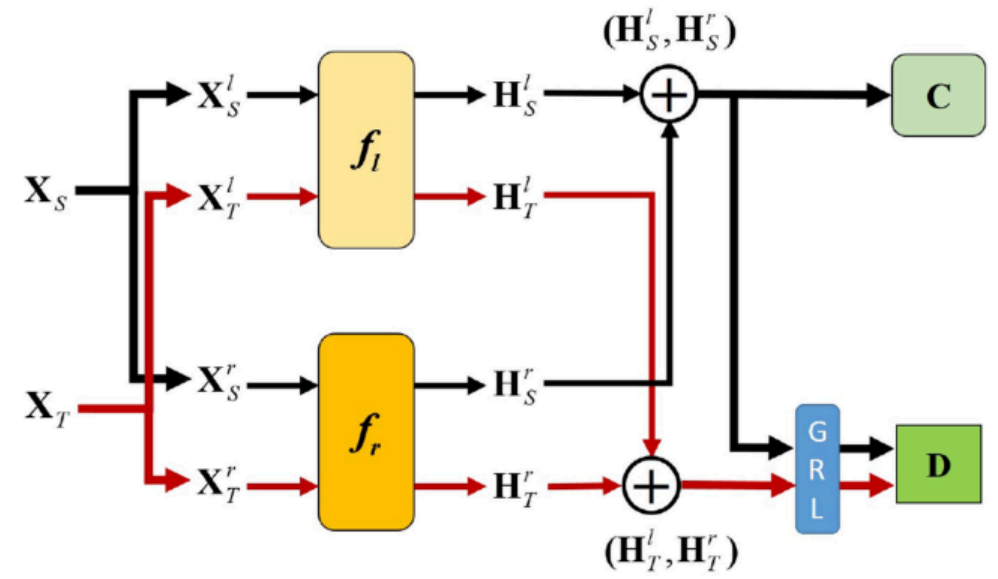


Fig. 5. The framework of BiDANN-R2, which uses two feature extractors to capture each brain hemispheric emotion information from source and target data separately.

The frameworks of BiDANN-R1 and BiDANN-R2 are shown in Figs. 4 and 5, respectively.

5.1 Subject-Dependent EEG Emotion Recognition Experiments

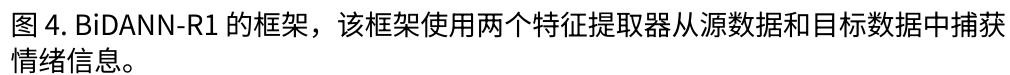
In this experiment, we adopt the experimental protocol of Zheng et al. [13] by using 9 trials of EEG data per session of each subject as source (training) domain data whereas using the remaining 6 trials per session as target (testing) domain data. In this way, we can obtain 1938 samples in total as the training data and 1336 samples in total as the testing data. For comparison purpose, we use the linear support vector machine (SVM) [37], canonical correlation analysis (CCA) [39], group sparse canonical correlation analysis [11], deep believe network [13], graph regularization sparse linear regression [12], graph convolutional neural network (GCNN) [40] and dynamical graph convolutional neural network (DGCNN) [14] to conduct the same experiment, respectively. Moreover, in this experiment we also conduct additional experiments for the BiDANN-R1 method by using hemispheric EEG data to investigate which hemisphere contributes more to the emotion recognition. Table 2 summarizes the experimental results of the various methods. In addition, we also show the confusion matrices of the BiDANN method in Fig. 6 to see the confusions of BiDANN in recognizing the three emotions.

From Table 2, we can see that the proposed BiDANN method achieves the best recognition result among the various methods. Especially, compared with the other deep learning based methods such as DBN, GCNN and DGCNN, the proposed BiDANN method increases the classification accuracy of 6.30, 4.98 and 1.98 percent, respectively. In

TABLE 2
The Mean Accuracies (ACC) and Standard Deviations (STD) on SEED Database for Conventional EEG Emotion Recognition Experiment

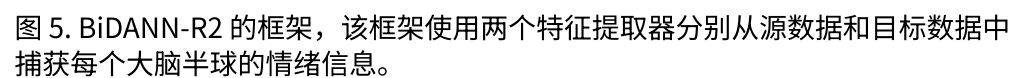
Method	SVM [37]	RF* [38]	CCA* [39]	GSCCA* [11]	DBN [13]	GRSLR* [12]	GCNN [40]
ACC/STD(%)	83.99/09.72	78.46/11.77	77.63/13.21	82.96/09.95	86.08/08.34	87.39/08.64	87.40/09.20
Method	DGCNN [14]	DANN* [17]	BiDANN-R1	BiDANN-R2	BiDANN		
ACC/STD(%)	90.40/08.49	91.36/08.30	90.29/08.02	91.60/08.47	92.38/07.04		

* Denotes the experiment results obtained are based on our own implementation.



5 实验

此外，为了评估 BiDANN 不同部分对脑电图情绪识别的贡献，我们还提出了 BiDANN 的两个简化版本，分别记为 BiDANN-R1 和 BiDANN-R2，并在相同的实验设置下比较它们的识别性能。在 BiDANN-R1 模型中，我们使用两个特征提取器从源数据和目标数据中捕获情绪信息，而不考虑半球差异。与 BiDANN-R1 模型不同，BiDANN-R2 模型使用两个特征提取器分别从源数据和目标数据中捕获每个大脑半球的情绪信息。



BiDANN-R1 和 BiDANN-R2 的框架分别如图 4 和图 5 所示。

5.1 受试者依赖性脑电图情绪识别实验

在本实验中，我们采用 Zheng 等人[13]的实验方案，将每个受试者每场实验的 9 条 EEG 数据作为源（训练）域数据，而将剩余的 6 条数据作为目标（测试）域数据。通过这种方式，我们总共获得了 1938 个样本作为训练数据，以及 1336 个样本作为测试数据。为了进行比较，我们分别使用线性支持向量机（SVM）[37]、典型相关分析（CCA）[39]、分组稀疏典型相关分析[11]、深度信念网络[13]、图正则化稀疏线性回归[12]、图卷积神经网络（GCNN）[40]和动态图卷积神经网络（DGCNN）[14]进行相同的实验。此外，在本实验中，我们还对 BiDANN-R1 方法进行了额外的实验，使用半球 EEG 数据来研究哪个半球对情绪识别贡献更大。表 2 总结了各种方法的实验结果。此外，我们在图 6 中展示了 BiDANN 方法的混淆矩阵，以观察 BiDANN 在识别三种情绪时的混淆情况。

从表 2 可以看出，所提出的 BiDANN 方法在多种方法中取得了最佳识别结果。特别是，与其他基于深度学习的方法（如 DBN、GCNN 和 DGCNN）相比，所提出的 BiDANN 方法分别提高了分类准确率 6.30%、4.98%和 1.98%。在

表 2. 在 SEED 数据库上传统脑电图情绪识别实验的平均准确率 (ACC) 和标准差 (STD)

方法 SVM [37] RF[38] CCA[39] GSCCA[11] DBN [13] GRSLR[12] GCNN [40]					
ACC/STD(%)	83.99/09.72	78.46/11.77	77.63/13.21	82.96/09.95	86.08/08.34 87.39/08.64 87.40/09.20
方法 DGCNN [14] DANN[17] BiDANN-R1 BiDANN-R2 BiDANN					
ACC/STD(%)	90.40/08.49	91.36/08.30	90.29/08.02	91.60/08.47	92.38/07.04

表示实验结果基于我们自己的实现。

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negative	86.15	9.27	4.58
neutral	4.99	93.61	1.40
positive	1.42	1.69	96.89
	negative	neutral	positive

Fig. 6. The confusion matrices of the subject-dependent EEG emotion recognition results using the BiDANN method on the SEED database.

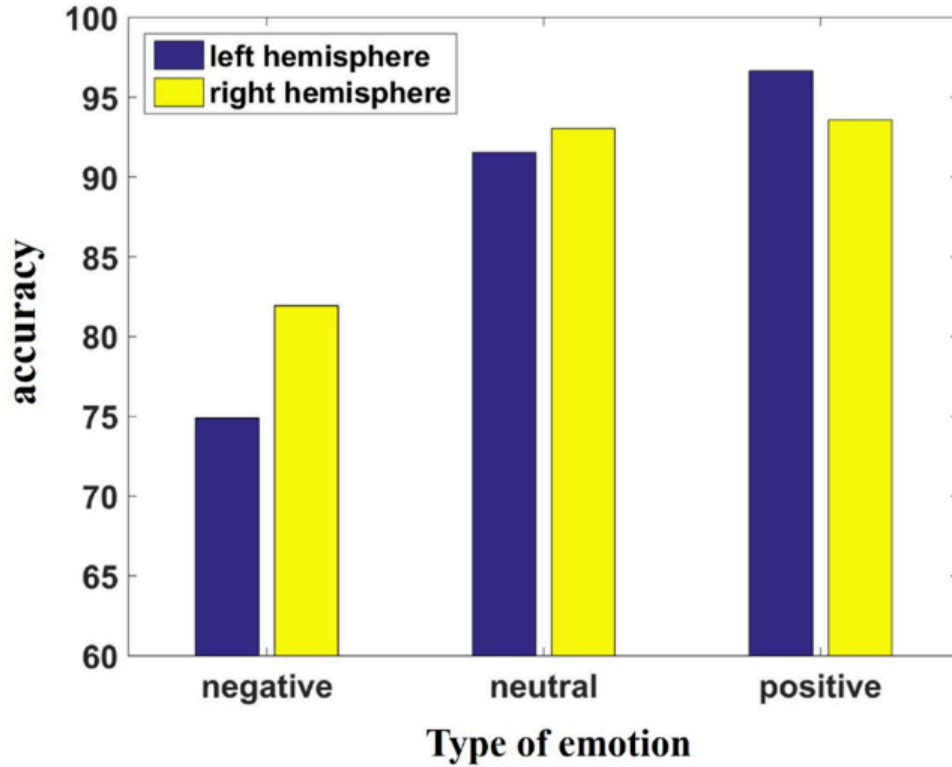


Fig. 7. The experimental results of using the BiDANN-R1 method to recognizing the three emotions based on the left and the right hemispheres EEG signals, respectively.

addition, from Table 2, we can see that the methods with domain discriminator, such as DANN and BiDANN-R1, achieve higher recognition accuracies than the other methods. On the other hand, from Fig. 6, we can see that the average recognition accuracies of BiDANN in recognizing the three types of emotions are 86.15 (negative), 93.61 (neutral), and 96.89 percent (positive), which means that positive emotion would be much easier to be recognized than both negative and neutral emotions.

Moreover, we also adopt the BiDANN-R1 method to investigate the differences between left and right hemispheres in the perception of the three types of emotions. The experimental results are shown in Fig. 7, from which we can see that, the right hemisphere shows more powerful perception ability than the left hemisphere in perceiving the negative emotion, whereas the left hemisphere demonstrates better perception ability than the right hemisphere in perceiving the positive emotion. The experimental results are consistent with the neuroscience finding about the emotion perception of the two hemispheres [23].

It is notable that the previous work of Zheng et al. [11] showed that different frequency bands of the EEG signals may play different roles in the emotion recognition task. In this experiment, we also conduct experiments to verify this point. Table 3 summarizes the experimental results of the

TABLE 3

The Mean Accuracies (and Standard Deviations) Using Different Frequency Bands for Subject-Dependent EEG Emotion Recognition Experiment on SEED Database

Methods	Frequency bands				
	δ	θ	α	β	γ
SVM [37]	60.50 (14.14)	60.95 (10.20)	66.64 (14.41)	80.76 (11.56)	79.56 (11.38)
RF* [38]	64.56 (08.32)	65.27 (11.64)	65.67 (13.94)	73.35 (14.35)	74.48 (12.80)
CCA* [39]	55.30 (12.02)	55.75 (10.99)	64.96 (12.05)	69.16 (11.45)	70.67 (14.06)
GSCCA* [11]	63.92 (11.16)	64.64 (10.33)	70.10 (14.76)	76.93 (11.00)	77.98 (10.72)
DBN [13]	64.32 (12.45)	60.77 (10.42)	64.01 (15.97)	78.92 (12.48)	79.19 (14.58)
GRSLR* [12]	63.90 (11.83)	62.61 (10.73)	71.11 (09.04)	81.18 (10.74)	81.91 (10.36)
GCNN [40]	72.75 (10.85)	74.40 (08.23)	73.46 (12.17)	83.24 (09.93)	83.36 (09.43)
DGCNN [14]	74.25 (11.42)	71.52 (05.99)	74.43 (12.16)	83.65 (10.17)	85.73 (10.64)
DANN* [17]	72.13 (11.22)	68.75 (07.40)	70.27 (10.84)	83.35 (11.46)	87.89 (11.35)
BiDANN-R1	71.63 (10.69)	66.79 (08.96)	67.71 (12.49)	85.68 (10.13)	86.93 (09.56)
BiDANN-R2	75.26 (10.95)	71.20 (07.73)	75.40 (14.26)	87.73 (10.58)	86.80 (10.78)
BiDANN	76.97 (10.95)	75.56 (07.88)	81.03 (11.74)	89.65 (09.59)	88.64 (09.46)

* Denotes the experiment results obtained are based on our own implementation.

various emotion recognition methods across the five EEG frequency bands. From Table 3, we can see that the experimental results are consistent with the findings of Zheng et al. in [11], i.e., the higher frequency bands (β and γ) would achieve better recognition performance than lower frequency bands (δ , θ and α). From Table 3, we can also see that the BiDANN method achieves better performance than the other methods for each frequency band.

5.2 Subject-Independent EEG Emotion Recognition Experiments

In this experiment, we will focus our attention on the subject-independent EEG emotion recognition problem. For this purpose, we adopt the leave-one-subject-out (LOSO) cross-validation strategy to evaluate the performance of the proposed BiDANN-S method, in which we use the EEG signals of one subject as testing data and the rest of ones of 14 subjects as training data to conduct the experiment. This procedure is repeated such that the EEG signals of each subject have been used once as testing data. For comparison purpose, we also conduct the same experiments using various methods, including the methods of Kullback-Leibler importance estimation procedure (KLIEP) [41], unconstrained least-squares importance fitting (ULSIF) [42], selective transfer machine (STM) [43], linear SVM [37], kernel principal component analysis (KPCA) [44], transfer component analysis (TCA) [45], transfer kernel learning (TKL) [46], subspace alignment (SA) [47], geodesic flow kernel (GFK) [48], transductive SVM

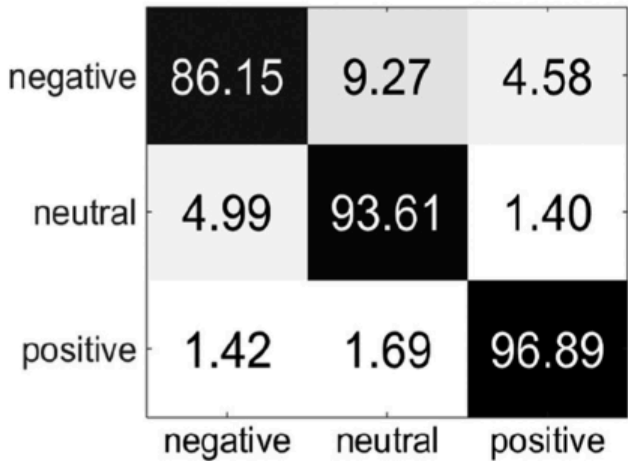


图 6. 在 SEED 数据库上使用 BiDANN 方法进行受试者相关脑电图情绪识别结果的混淆矩阵。

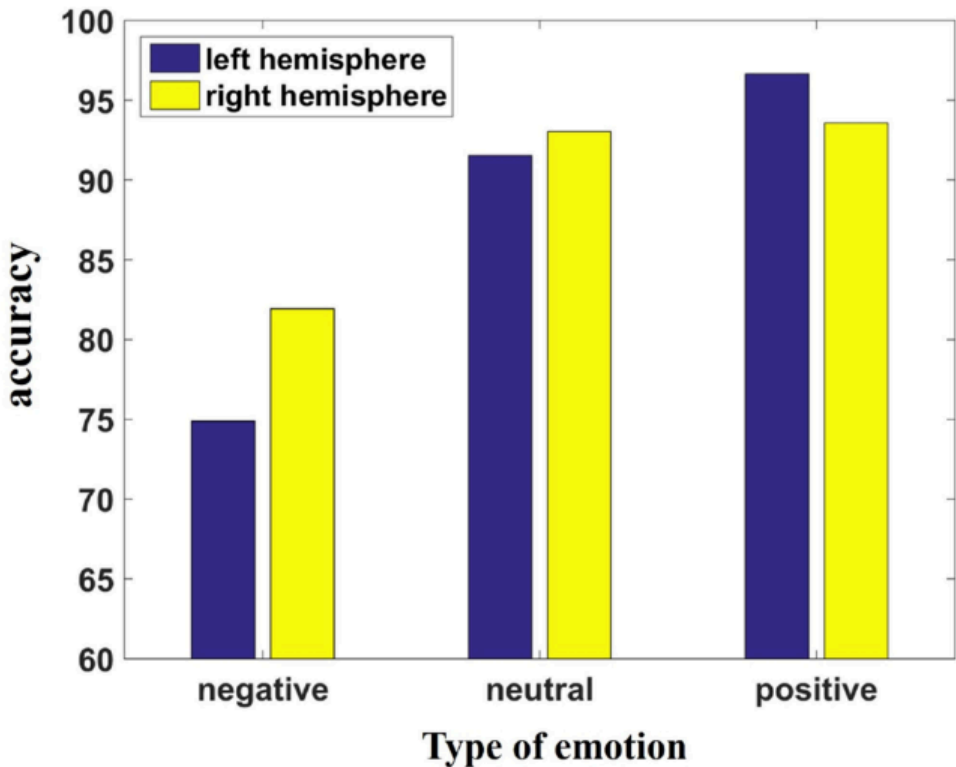


图 7. 使用 BiDANN-R1 方法分别基于左右半球脑电信号识别三种情绪的实验结果。

此外，从表 2 可以看出，具有域判别器的方法，如 DANN 和 BiDANN-R1，比其他方法的识别准确率更高。另一方面，从图 6 可以看出，BiDANN 识别三种情绪的平均识别准确率分别为 86.15%（负面）、93.61%（中性）和 96.89%（正面），这意味着与负面和中性情绪相比，正面情绪更容易被识别。

此外，我们还采用 BiDANN-R1 方法来研究左右半球在感知三种类型情绪时的差异。实验结果如图 7 所示，从中我们可以看出，右半球在感知负面情绪时表现出比左半球更强的感知能力，而左半球在感知正面情绪时则显示出比右半球更好的感知能力。实验结果与关于两半球情绪感知的神经科学发现[23]一致。

值得注意的是，Zheng 等人[11]的先前研究表明，脑电信号的不同频段在情绪识别任务中可能扮演不同的角色。在本实验中，我们也进行了实验来验证这一点。

表 3. 用于受试者相关脑电情绪识别的不同频段平均准确率（及标准差）。

SEED 数据库上的识别实验					
方法	频段				
	d	u	a	b	g
SVM [37]	60.50 (14.14)	60.95 (10.20)	66.64 (14.41)	80.76 (11.56)	79.56 (11.38)
RF[38]	64.56 (08.32)	65.27 (11.64)	65.67 (13.94)	73.35 (14.35)	74.48 (12.80)
CCA[39]	55.30 (12.02)	55.75 (10.99)	64.96 (12.05)	69.16 (11.45)	70.67 (14.06)
GSCCA[11]	63.92 (11.16)	64.64 (10.33)	70.10 (14.76)	76.93 (11.00)	77.98 (10.72)
DBN [13]	64.32 (12.45)	60.77 (10.42)	64.01 (15.97)	78.92 (12.48)	79.19 (14.58)
GRSLR[12]	63.90 (11.83)	62.61 (10.73)	71.11 (09.04)	81.18 (10.74)	81.91 (10.36)
GCNN [40]	72.75 (10.85)	74.40 (08.23)	73.46 (12.17)	83.24 (09.93)	83.36 (09.43)
DGCNN [14]	74.25 (11.42)	71.52 (05.99)	74.43 (12.16)	83.65 (10.17)	85.73 (10.64)
DANN[17]	72.13 (11.22)	68.75 (07.40)	70.27 (10.84)	83.35 (11.46)	87.89 (11.35)
BiDANN-R1	71.63 (10.69)	66.79 (08.96)	67.71 (12.49)	85.68 (10.13)	86.93 (09.56)
BiDANN-R2	75.26 (10.95)	71.20 (07.73)	75.40 (14.26)	87.73 (10.58)	86.80 (10.78)
BiDANN	76.97 (10.95)	75.56 (07.88)	81.03 (11.74)	89.65 (09.59)	88.64 (09.46)

表示实验结果基于我们自己的实现。

表 3 总结了在五个脑电图频段上各种情绪识别方法的实验结果。从表 3 中可以看出，实验结果与 Zheng 等人[11]的研究发现一致，即高频段（ β 和 γ ）的识别性能优于低频段（ δ 、 θ 和 α ）。从表 3 中还可以看出，BiDANN 方法在每个频段上都取得了比其他方法更好的性能。

5.2 独立于被试的脑电图情绪识别实验

在这个实验中，我们将重点关注受试者无关的脑电图情绪识别问题。为此，我们采用留一受试者交叉验证（LOSO）策略来评估所提出的 BiDANN-S 方法的性能，其中我们使用一个受试者的脑电图信号作为测试数据，其余 14 个受试者的信号作为训练数据来开展实验。该过程重复进行，直到每个受试者的脑电图信号都作为测试数据使用过一次。为了比较，我们还使用各种方法进行了相同的实验，包括 Kullback-Leibler 重要性估计程序（KLIEP）[41]、无约束最小二乘重要性拟合（ULSIF）[42]、选择性迁移机（STM）[43]、线性支持向量机（SVM）[37]、核主成分分析（KPCA）[44]、迁移成分分析（TCA）[45]、迁移核学习（TKL）[46]、子空间对齐（SA）[47]、测地流核（GFK）[48]、判别支持向量机

TABLE 4
The Mean Accuracies (ACC) and Standard Deviations (STD) on SEED Database for Subject-Independent EEG Emotion Recognition Experiment

Method	KLIEP* [41]	ULSIF* [42]	STM* [43]	SVM [37]	KPCA [44]	TCA [45]
ACC/STD(%)	45.71/17.76	51.18/13.57	51.23/14.82	56.73/16.29	61.28/14.62	63.64/14.88
Method	TKL* [46]	SA* [47]	GFK* [48]	T-SVM [49]	TPT [50]	DGCNN [14]
ACC/STD(%)	63.54/15.47	69.00/10.89	71.31/14.09	72.53/14.00	76.31/15.89	79.95/09.02
Method	DANN* [17]	BiDANN-R1	BiDANN-R2	BiDANN	BiDANN-S	
ACC/STD(%)	75.08/11.18	76.97/11.08	82.22/07.61	83.28/09.60	84.14/06.87	

* Denotes the experiment results obtained are based on our own implementation.

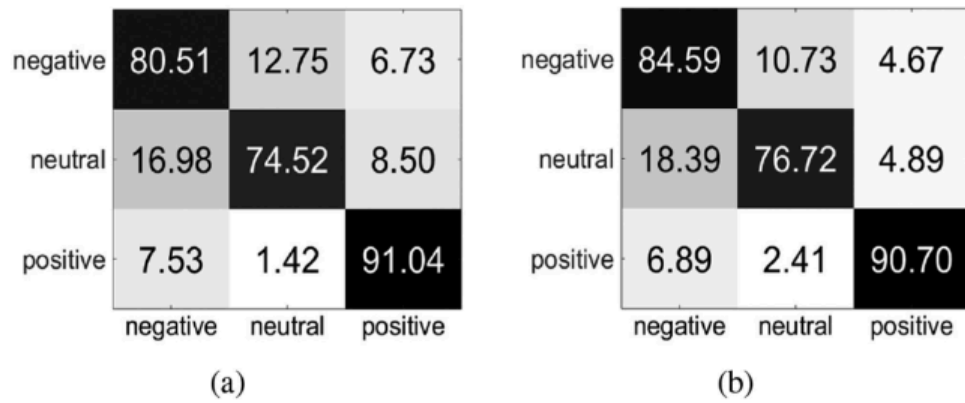


Fig. 8. The confusion matrices of the subject-independent EEG emotion recognition results using BiDANN and BiDANN-S method on the SEED database. (a) Confusion matrix of BiDANN; (b) Confusion matrix of BiDANN-S.

(T-SVM) [49], transductive parameter transfer (TPT) [50], and DGCNN [14]. The experimental results of the various methods are shown in Table 4.¹

From Table 4, we can see that the proposed BiDANN-S method achieves the best recognition result among the various methods. The better recognition performance of BiDANN-S is very likely due to the fact that the subject discriminator of BiDANN-S can effectively reduce the subjects' personal identity information such that it is more powerful for the subject-independent emotion recognition problem. Moreover, we can also see that the BiDANN-R2 method achieves better recognition result than the BiDANN-R1 method, which indicates the importance of considering the discrepancy information between left and right cerebral hemispheric data in the EEG emotion recognition. Furthermore, we can also see that the BiDANN method with local discriminators also achieves better performance than BiDANN-R2, which indicates that the local discriminators are useful to improve the EEG emotion recognition performance.

In addition, we also depict the confusion matrices corresponding to the experimental results of BiDANN and BiDANN-S to see the confusions of the three emotions. Fig. 8 shows the confusion matrices of the experimental results of BiDANN and BiDANN-S, respectively. From Fig. 8, we can see that both BiDANN and BiDANN-S perform well in recognizing all the three types of emotion. Compared with BiDANN, it is interesting to see that BiDANN-S perform

1. Noting that the subspace based methods, such as KLIEP, ULSIF, STM, TCA, TKL, SA, GFK and KPCA, are problematic to handle larger size of EEG data due to the of computer memory limitation and computational problem, we only randomly select 5000 EEG feature samples from the training data set to train these methods.

much better in recognizing both negative and neutral emotions than BiDANN, indicating the effectiveness of the subject discriminator in the BiDANN-S model.

Finally, to investigate the different contributions of the different frequency bands to the emotion recognition, we also conduct the experiments on each of the five frequency bands using the LOSO experimental strategy. Table 5 summarizes the experimental results of the various methods with respect to the five frequency bands. Again, from Table 5, we can see that the higher frequency bands achieve

TABLE 5
The Mean Accuracies (and Standard Deviations) Using Different Frequency Bands for Subject-Independent EEG Emotion Recognition on SEED Database

Methods	Frequency bands				
	δ	θ	α	β	γ
KLIEP* [41]	39.22 (11.31)	35.98 (07.50)	33.31 (06.60)	44.47 (12.89)	42.05 (12.65)
ULSIF* [42]	41.32 (11.30)	36.27 (06.84)	38.94 (08.30)	41.87 (13.64)	41.02 (11.65)
STM* [43]	44.16 (09.60)	40.89 (08.22)	40.37 (09.82)	42.09 (13.34)	47.97 (12.43)
SVM* [37]	43.06 (08.27)	40.07 (06.50)	43.97 (10.89)	48.63 (10.29)	51.59 (11.83)
TCA* [45]	44.10 (08.22)	41.26 (09.21)	42.93 (14.33)	43.93 (10.06)	48.43 (09.73)
TKL* [46]	48.36 (10.31)	52.60 (11.84)	52.89 (11.07)	55.47 (09.80)	59.81 (12.41)
SA* [47]	53.23 (07.47)	50.60 (08.31)	55.06 (10.60)	56.72 (10.78)	64.47 (14.96)
GFK* [48]	52.73 (11.90)	54.07 (06.78)	54.98 (11.49)	59.29 (10.75)	66.92 (10.97)
DGCNN [14]	49.79 (10.94)	46.36 (12.06)	48.29 (12.28)	56.15 (14.01)	54.87 (17.53)
DANN* [17]	56.66 (06.48)	54.95 (10.45)	59.37 (10.57)	67.14 (07.10)	71.30 (10.84)
BiDANN-R1	57.33 (06.76)	57.00 (08.92)	58.20 (13.50)	64.76 (13.79)	65.15 (14.14)
BiDANN-R2	59.67 (10.48)	60.70 (07.42)	61.08 (10.77)	74.09 (11.54)	72.77 (11.51)
BiDANN	62.04 (06.64)	62.13 (07.37)	63.31 (11.46)	73.55 (08.83)	73.25 (09.21)
BiDANN-S	63.01 (07.49)	63.22 (07.52)	63.50 (09.50)	73.59 (09.12)	73.72 (08.67)

* Denotes the experiment results obtained are based on our own implementation.

表 4 SEED 数据库上主题独立脑电情绪识别实验的平均准确率（ACC）和标准差（STD）

方法	KLIEP[41]	ULSIF[42]	STM[43]	SVM [37]	KPCA [44]	TCA [45]
ACC/STD(%)	45.71/17.76	51.18/13.57	51.23/14.82	56.73/16.29	61.28/14.62	63.64/14.88
方法	TKL[46]	SA[47]	GFK[48]	T-SVM [49]	TPT [50]	DGCNN [14]
ACC/STD(%)	63.54/15.47	69.00/10.89	71.31/14.09	72.53/14.00	76.31/15.89	79.95/09.02
方法	DANN[17] BiDANN-R1 BiDANN-R2 BiDANN BiDANN-S					
ACC/STD(%)	75.08/11.18	1	76.97/11.08	82.22/07.61	83.28/09.60	84.14/06.87

表示实验结果基于我们自己的实现。

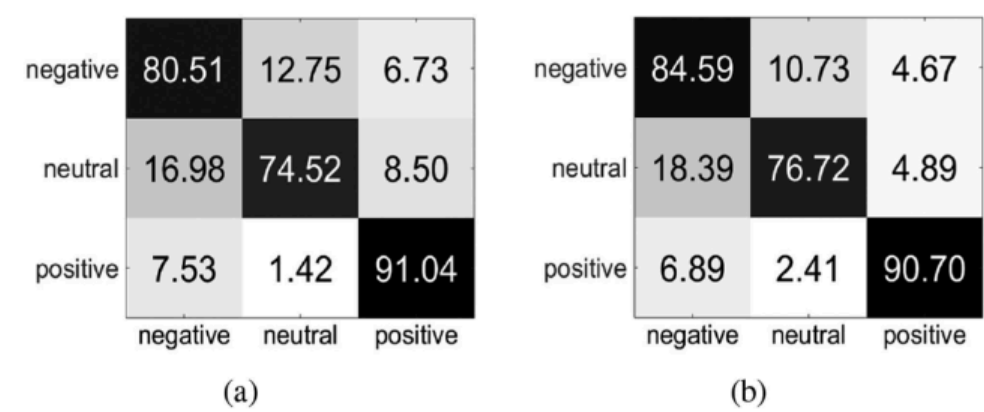


图 8. 在 SEED 数据库上使用 BiDANN 和 BiDANN-S 方法进行主题无关的脑电图情绪识别结果的混淆矩阵。(a) BiDANN 的混淆矩阵；(b) BiDANN-S 的混淆矩阵。

(T-SVM) [49], 传递参数迁移 (TPT) [50], 和 DGCNN [14]。各种方法的实验结果如表 4 所示。

从表 4 中可以看出，所提出的 BiDANN-S 方法在各种方法中取得了最佳识别结果。BiDANN-S 识别性能的更好可能是由于它的被试判别器能够有效减少被试的个人身份信息，从而使其在主体无关的情绪识别问题上更具优势。此外，我们还可以看到 BiDANN-R2 方法的识别结果优于 BiDANN-R1 方法，这表明在脑电图情绪识别中考虑左右脑半球数据差异的重要性。此外，我们还可以看到具有局部判别器的 BiDANN 方法也取得了比 BiDANN-R2 更好的性能，这表明局部判别器对于提高脑电图情绪识别性能是有用的。

此外，我们还展示了 BiDANN 和 BiDANN-S 的实验结果对应的混淆矩阵，以观察三种情绪的混淆情况。图 8 分别展示了 BiDANN 和 BiDANN-S 的实验结果混淆矩阵。从图 8 可以看出，BiDANN 和 BiDANN-S 在识别所有三种类型的情绪方面表现良好。与 BiDANN 相比

需要注意的是，基于子空间的方法，如 KLIEP、ULSIF、STM、TCA、TKL、SA、GFK 和 KPCA，由于计算机内存限制和计算问题，难以处理较大规模的脑电图数据，因此我们仅从训练数据集中随机选择 5000 个脑电图特征样本来训练这些方法。

有趣的是，BiDANN-S 在识别负性和中性情绪方面表现远好于 BiDANN，这表明 BiDANN-S 模型中的主题判别器是有效的。

最后，为了研究不同频段对情绪识别的不同贡献，我们使用 LOSO 实验策略对每个五个频段分别进行实验。表 5 总结了各种方法在五个频段上的实验结果。再次从表 5 可以看出，高频段实现了

表 5 使用不同频段对受试者无关的脑电图情绪识别的平均准确率（和标准差）

SEED 数据库上的识别					
方法	频段				
	d	u	a	b	g
KLIEP[41]	39.22	35.98	33.31	44.47	42.05
	(11.31)	(07.50)	(06.60)	(12.89)	(12.65)
ULSIF[42]	41.32	36.27	38.94	41.87	41.02
	(11.30)	(06.84)	(08.30)	(13.64)	(11.65)
STM[43]	44.16	40.89	40.37	42.09	47.97
	(09.60)	(08.22)	(09.82)	(13.34)	(12.43)
SVM[37]	43.06	40.07	43.97	48.63	51.59
	(08.27)	(06.50)	(10.89)	(10.29)	(11.83)
TCA[45]	44.10	41.26	42.93	43.93	48.43
	(08.22)	(09.21)	(14.33)	(10.06)	(09.73)
TKL[46]	48.36	52.60	52.89	55.47	59.81
	(10.31)	(11.84)	(11.07)	(09.80)	(12.41)
SA[47]	53.23	50.60	55.06	56.72	64.47
	(07.47)	(08.31)	(10.60)	(10.78)	(14.96)
GFK[48]	52.73	54.07	54.98	59.29	66.92
	(11.90)	(06.78)	(11.49)	(10.75)	(10.97)
DGCNN [14]	49.79	46.36	48.29	56.15	54.87
	(10.94)	(12.06)	(12.28)	(14.01)	(17.53)
DANN[17]	56.66	54.95	59.37	67.14	71.30
	(06.48)	(10.45)	(10.57)	(07.10)	(10.84)
BiDANN-R1	57.33	57.00	58.20	64.76	65.15
	(06.76)	(08.92)	(13.50)	(13.79)	(14.14)
BiDANN-R2	59.67	60.70	61.08	74.09	72.77
	(10.48)	(07.42)	(10.77)	(11.54)	(11.51)
BiDANN	62.04	62.13	63.31	73.55	73.25
	(06.64)	(07.37)	(11.46)	(08.83)	(09.21)
BiDANN-S	63.01	63.22	63.50	73.59	73.72
	(07.49)	(07.52)	(09.50)	(09.12)	(08.67)

实验结果基于我们自己的实现获得。

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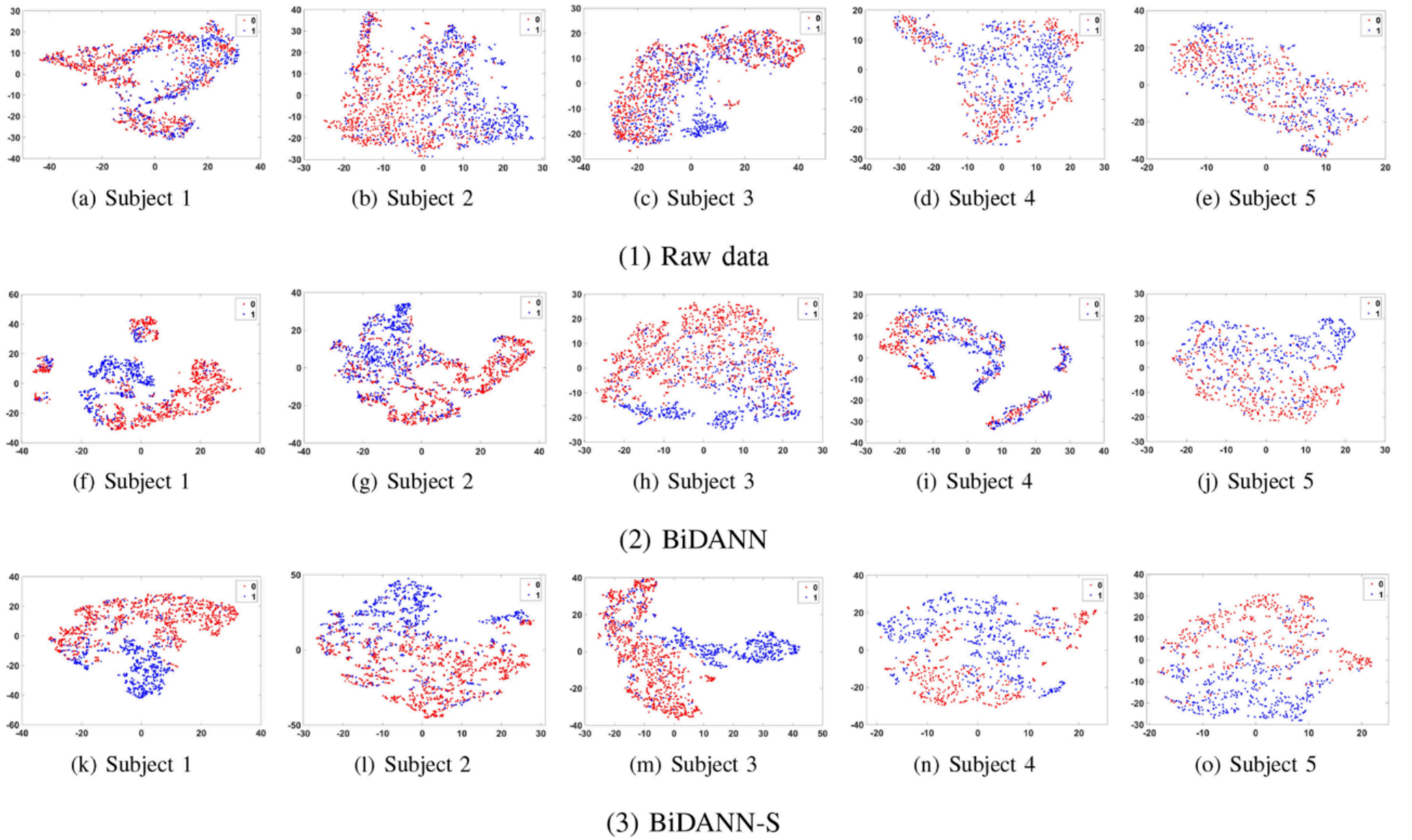


Fig. 9. The t-SNE visualization before and after domain adaptation. (a)-(e) are raw data. (f)-(j) are feature distributions obtained by BiDANN. (k)-(o) are feature distributions obtained by BiDANN-S. The red color denotes the negative emotion and the blue color denotes the positive emotion.

better recognition performance than lower frequency bands. Moreover, from Table 5 we can also see that the proposed BiDANN and BiDANN-S methods achieve better recognition performance than the other methods for the five frequency bands.

5.3 Discussions

In the aforementioned experiments, we had demonstrated that the BiDANN-S method can further improve the recognition performance in contrast to the BiDANN method. This mainly attributes to the fact that the subject discriminator used in BiDANN-S can largely remove the influences of the subject information from the EEG features and hence is able to improve the discriminative ability of the EEG features for emotion recognition.

To further verify this point, we conduct an additional experiment on the two-class HR-EEG4EMO database [51], which consists of the EEG signals of 27 subjects. In this experiment, we first adopt the same band power (BP) EEG feature as that used in [51] to extract 64 handcraft features associated with 64 EEG electrodes from each subject. Then, we visualize the feature distributions associated with five subjects using the t-SNE embeddings method [52]. Fig. 9 shows the visualized results of the raw EEG handcraft features and the discriminative features learnt by BiDANN and BiDANN-S respectively, where the feature distribution corresponding to each subject is obtained based on BiDANN or BiDANN-S using the EEG data of the other 26 subjects. From Fig. 9, we can see that the raw EEG feature points associated with positive emotion are relatively hard to be recognized compared with the feature points

associated with negative emotion. In contrast to the raw features, we can see that the separability between the positive emotion and negative emotion becomes much easier when BiDANN or BiDANN-S is used. On the other hand, compared with the BiDANN method, the BiDANN-S method achieves better separability for most cases. The better separability of BiDANN-S than BiDANN mainly attributes to the reduction of the subject personal information in the EEG features.

6 CONCLUSIONS

In this paper, we had proposed a novel BiDANN method to deal with EEG emotion recognition, which is inspired by the neuroscience findings about the asymmetry property of human brain to the emotion response. The extensive experiments on the SEED EEG database demonstrated that the proposed BiDANN method achieves better EEG emotion recognition performance than the state-of-the-art methods. The better recognition performance of BiDANN mostly attributes to the fact that the BiDANN method can not only make use of the advantages of adversarial discriminative domain adaptation to improve the EEG recognition performance, but also is able to make use of the asymmetry of the left and right hemispheres in the perception of different emotions. Based on the BiDANN method, we also proposed the BiDANN-S method to further improve the BiDANN method to deal with the cross subject EEG emotion recognition problem. The experimental results on the SEED database had confirmed that the BiDANN-S method would achieve higher recognition accuracy than BiDANN.

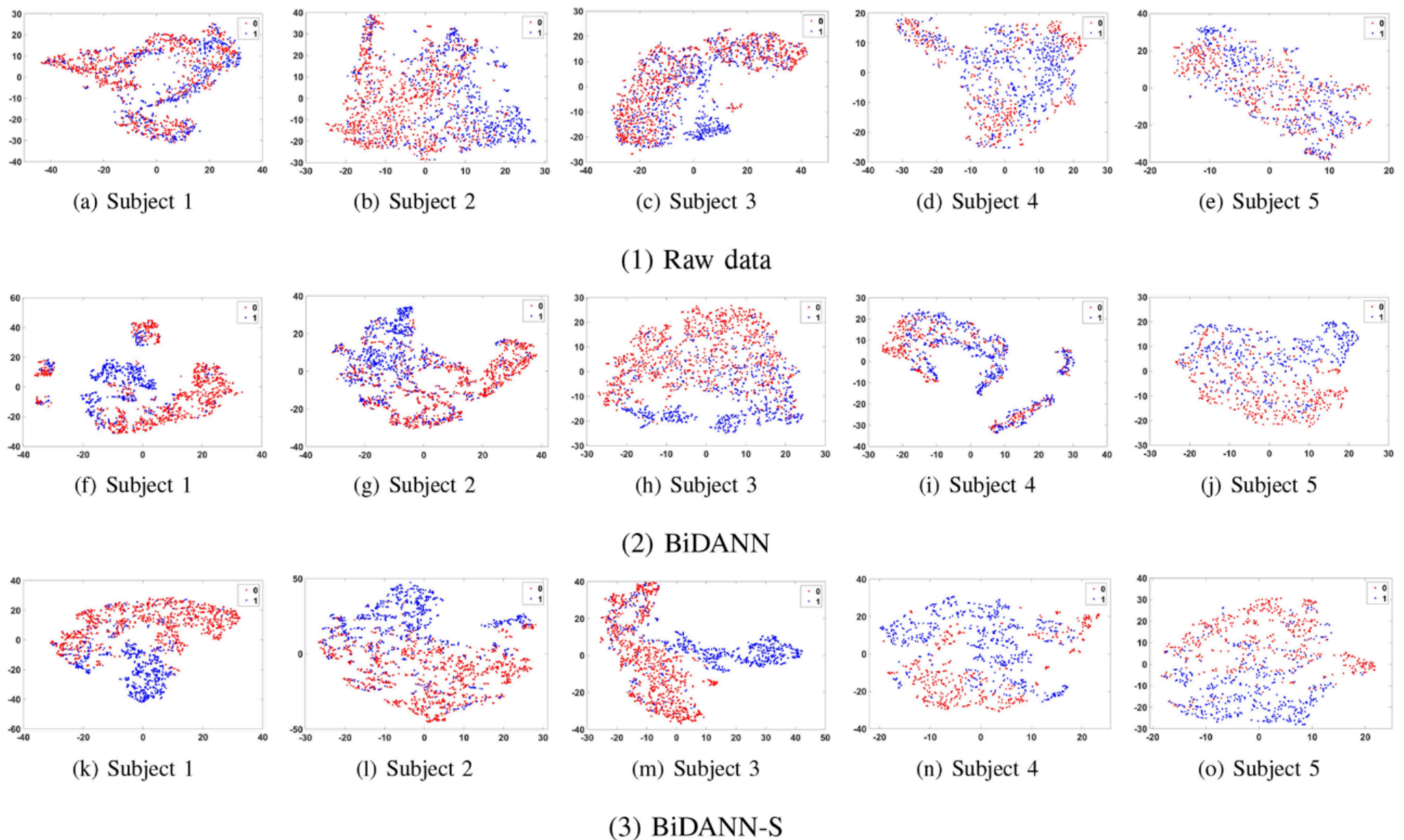


图 9. 域适应前后的 t-SNE 可视化。(a)-(e) 是原始数据。(f)-(j) 是 BiDANN 获得的特征分布。(k)-(o) 是 BiDANN-S 获得的特征分布。红色表示负面情绪，蓝色表示正面情绪。

比低频段具有更好的识别性能。此外，从表 5 中我们还可以看到，所提出的 BiDANN 和 BiDANN-S 方法在五个频段上均实现了比其他方法更好的识别性能。

5.3 讨论

在前述实验中，我们已经证明与 BiDANN 方法相比，BiDANN-S 方法能够进一步提高识别性能。这主要归因于 BiDANN-S 中使用的被试判别器能够大幅消除 EEG 特征中受被试信息的影响，从而提高了 EEG 特征在情绪识别中的判别能力。

为了进一步验证这一点，我们在包含 27 名受试者脑电信号的二分类 HR-EEG4EMO 数据库[51]上进行了额外实验。在该实验中，我们首先采用与[51]中相同的频带功率 (BP) 脑电特征，从每位受试者提取与 64 个脑电电极相关的 64 个手工特征。然后，我们使用 t-SNE 嵌入方法[52]对五个受试者的特征分布进行可视化。图 9 展示了原始脑电手工特征和 BiDANN 及 BiDANN-S 分别学习到的判别特征的可视化结果，其中每个受试者的特征分布是基于其他 26 名受试者的脑电数据，使用 BiDANN 或 BiDANN-S 获得的。从图 9

我们可以看到，与负情绪相关的原始脑电图特征点相比，与正情绪相关的原始脑电图特征点相对较难识别。与原始特征相比，当使用 BiDANN 或 BiDANN-S 时，正情绪与负情绪之间的可分性变得容易得多。另一方面，与 BiDANN 方法相比，BiDANN-S 方法在大多数情况下实现了更好的可分性。BiDANN-S 比 BiDANN 具有更好的可分性主要归因于脑电图特征中受试者个人信息减少。

6 结论

在本文中，我们提出了一种新的 BiDANN 方法来处理脑电图情绪识别问题，该方法受到神经科学关于人类大脑对情绪反应的不对称性发现的启发。在 SEED 脑电图数据库上的大量实验表明，所提出的 BiDANN 方法比现有最先进方法实现了更好的脑电图情绪识别性能。BiDANN 更好的识别性能主要归因于该方法不仅能够利用对抗判别域适应的优势来提高脑电图识别性能，而且还能利用左右半球在感知不同情绪时的不对称性。基于 BiDANN 方法，我们还提出了 BiDANN-S 方法，以进一步改进 BiDANN 方法来处理跨被试的脑电图情绪识别问题。在 SEED 数据库上的实验结果已经证实，BiDANN-S 方法将比 BiDANN 实现更高的识别准确率。

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REFERENCES

- [1] R. J. Dolan, "Emotion, cognition, and behavior," *Sci.*, vol. 298, no. 5596, pp. 1191–1194, 2002.
- [2] R. W. Picard and R. Picard, *Affective Computing*, vol. 252. Cambridge, MA, USA: MIT Press, 1997.
- [3] C. Mühl, B. Allison, A. Nijholt, and G. Chanel, "A survey of affective brain computer interfaces: Principles, state-of-the-art, and challenges," *Brain-Comput. Interfaces*, vol. 1, no. 2, pp. 66–84, 2014.
- [4] E. Lotfi and M.-R. Akbarzadeh-T, "Practical emotional neural networks," *Neural Netw.*, vol. 59, pp. 61–72, 2014.
- [5] J. C. Britton, K. L. Phan, S. F. Taylor, R. C. Welsh, K. C. Berridge, and I. Liberzon, "Neural correlates of social and nonsocial emotions: An fmri study," *Neuroimage*, vol. 31, no. 1, pp. 397–409, 2006.
- [6] A. Etkin, T. Egner, and R. Kalisch, "Emotional processing in anterior cingulate and medial prefrontal cortex," *Trends Cognitive Sci.*, vol. 15, no. 2, pp. 85–93, 2011.
- [7] K. A. Lindquist and L. F. Barrett, "A functional architecture of the human brain: Emerging insights from the science of emotion," *Trends Cognitive Sci.*, vol. 16, no. 11, pp. 533–540, 2012.
- [8] R. Jenke, A. Peer, and M. Buss, "Feature extraction and selection for emotion recognition from eeg," *IEEE Trans. Affective Comput.*, vol. 5, no. 3, pp. 327–339, Jul.-Sep. 2014.
- [9] T. Musha, Y. Terasaki, H. A. Haque, and G. A. Ivamitsky, "Feature extraction from eegs associated with emotions," *Artificial Life Robot.*, vol. 1, no. 1, pp. 15–19, 1997.
- [10] M.-K. Kim, M. Kim, E. Oh, and S.-P. Kim, "A review on the computational methods for emotional state estimation from the human eeg," *Comput. Math. Methods Med.*, vol. 2013, 2013, Art. no. 573734.
- [11] W. Zheng, "Multichannel eeg-based emotion recognition via group sparse canonical correlation analysis," *IEEE Trans. Cognitive Develop. Syst.*, vol. 9, no. 3, pp. 281–290, Sep. 2017.
- [12] Y. Li, W. Zheng, Z. Cui, Y. Zong, and S. Ge, "EEG emotion recognition based on graph regularized sparse linear regression," *Neural Process. Lett.*, pp. 1–17, 2018, doi: <https://doi.org/10.1007/s11063-018-9829-1>.
- [13] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for eeg-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Mental Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.
- [14] T. Song, W. Zheng, P. Song, and Z. Cui, "EEG emotion recognition using dynamical graph convolutional neural networks," *IEEE Trans. Affective Comput.*, 2018, doi: [10.1109/TAFFC.2018.2817622](https://doi.org/10.1109/TAFFC.2018.2817622).
- [15] P. Pandey and K. Seeja, "Emotional state recognition with eeg signals using subject independent approach," in *Data Science and Big Data Analytics*. New York, NY, USA: Springer, 2019, pp. 117–124.
- [16] X. Li, D. Song, P. Zhang, Y. Zhang, Y. Hou, and B. Hu, "Exploring eeg features in cross-subject emotion recognition," *Frontiers Neuroscience*, vol. 12, 2018, Art. no. 162.
- [17] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette, M. Marchand, and V. Lempitsky, "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 59, pp. 1–35, 2016.
- [18] Y. Li, W. Zheng, Z. Cui, T. Zhang, and Y. Zong, "A novel neural network model based on cerebral hemispheric asymmetry for eeg emotion recognition," in *Proc. 27th Int. Joint Conf. Artif. Intell.*, 2018, pp. 1561–1567.
- [19] D. M. Tucker, "Lateral brain function, emotion, and conceptualization," *Psychological Bulletin*, vol. 89, no. 1, 1981, Art. no. 19.
- [20] N. Geschwind and A. M. Galaburda, "Cerebral lateralization: Biological mechanisms, associations, and pathology: I. a hypothesis and a program for research," *Archives Neurology*, vol. 42, no. 5, pp. 428–459, 1985.
- [21] R. J. Zatorre, M. Jones-Gotman, A. C. Evans, and E. Meyer, "Functional localization and lateralization of human olfactory cortex," *Nature*, vol. 360, no. 6402, pp. 339–340, 1992.
- [22] D. N. Greve, L. Van der Haegen, Q. Cai, S. Stufflebeam, M. R. Sabuncu, B. Fischl, and M. Brysbaert, "A surface-based analysis of language lateralization and cortical asymmetry," *J. Cognitive Neurosci.*, vol. 25, no. 9, pp. 1477–1492, 2013.
- [23] S. J. Dimond, L. Farrington, and P. Johnson, "Differing emotional response from right and left hemispheres," *Nature*, vol. 261, no. 5562, pp. 690–692, 1976.
- [24] R. J. Davidson, P. Ekman, C. D. Saron, J. A. Senulis, and W. V. Friesen, "Approach-withdrawal and cerebral asymmetry: Emotional expression and brain physiology: I," *J. Personality Social Psychology*, vol. 58, no. 2, 1990, Art. no. 330.
- [25] R. J. Davison, "Cerebral asymmetry, emotion, and affective style," in *Brain Asymmetry*, R. J. Davidson and K. Hugdahl Eds., Cambridge, MA, US: The MIT Press, 1995, pp. 361–387.
- [26] R. J. Davidson, D. Pizzagalli, J. B. Nitschke, and K. Putnam, "Depression: Perspectives from affective neuroscience," *Annu. Rev. Psychology*, vol. 53, no. 1, pp. 545–574, 2002.
- [27] E. Y. Costanzo, M. Villarreal, L. J. Drucaroff, M. Ortiz-Villafañe, M. N. Castro, M. Goldschmidt, A. E. Wainsztein, M. S. Ladrón-de Guevara, C. Romero, L. I. Brusco, et al., "Hemispheric specialization in affective responses, cerebral dominance for language, and handedness: Lateralization of emotion, language, and dexterity," *Behavioural Brain Res.*, vol. 288, pp. 11–19, 2015.
- [28] M. J. van Putten, "The revised brain symmetry index," *Clinical Neurophysiology*, vol. 118, no. 11, pp. 2362–2367, 2007.
- [29] H. Hinrikus, A. Suhhova, M. Bachmann, K. Aadamsoo, Ü. Võhma, J. Lass, and V. Tuulik, "Electroencephalographic spectral asymmetry index for detection of depression," *Med. Biological Eng. Comput.*, vol. 47, no. 12, 2009, Art. no. 1291.
- [30] J. M. Peters, M. Tomas-Fernandez, M. J. van Putten, and T. Loddenkemper, "Behavioral measures and eeg monitoring using the brain symmetry index during the wada test in children," *Epilepsy Behavior*, vol. 23, no. 3, pp. 247–253, 2012.
- [31] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [32] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative adversarial nets," in *Proc. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.
- [33] A. Jain, A. Singh, H. S. Koppula, S. Soh, and A. Saxena, "Recurrent neural networks for driver activity anticipation via sensory-fusion architecture," in *Proc. IEEE Int. Conf. Robot. Autom.*, 2016, pp. 3118–3125.
- [34] W. Wang, Z. Cui, Y. Yan, J. Feng, S. Yan, X. Shu, and N. Sebe, "Recurrent face aging," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 2378–2386.
- [35] L. Bottou, "Large-scale machine learning with stochastic gradient descent," in *Proc. COMPSTAT*, 2010, pp. 177–186.
- [36] W.-L. Zheng and B.-L. Lu, "Personalizing eeg-based affective models with transfer learning," in *Proc. 25th Int. Joint Conf. Artif. Intell.*, 2016, pp. 2732–2738.
- [37] J. A. Suykens and J. Vandewalle, "Least squares support vector machine classifiers," *Neural Process. Lett.*, vol. 9, no. 3, pp. 293–300, 1999.
- [38] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [39] B. Thompson, "Canonical correlation analysis," *Encyclopedia Statist. Behavioral Sci.*, Wiley Online Library, 2005.
- [40] M. Defferrard, X. Bresson, and P. Vandergheynst, "Convolutional neural networks on graphs with fast localized spectral filtering," in *Proc. Neural Inf. Process. Syst.*, 2016, pp. 3844–3852.
- [41] M. Sugiyama, S. Nakajima, H. Kashima, P. V. Buenau, and M. Kawanabe, "Direct importance estimation with model selection and its application to covariate shift adaptation," in *Proc. Neural Inf. Process. Syst.*, 2008, pp. 1433–1440.
- [42] T. Kanamori, S. Hido, and M. Sugiyama, "A least-squares approach to direct importance estimation," *J. Mach. Learn. Res.*, vol. 10, pp. 1391–1445, 2009.
- [43] W.-S. Chu, F. De la Torre, and J. F. Cohn, "Selective transfer machine for personalized facial expression analysis," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 3, pp. 529–545, Mar. 2017.
- [44] B. Schölkopf, A. Smola, and K.-R. Müller, "Nonlinear component analysis as a kernel eigenvalue problem," *Neural Comput.*, vol. 10, no. 5, pp. 1299–1319, 1998.
- [45] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, 2011.
- [46] M. Long, J. Wang, J. Sun, and S. Y. Philip, "Domain invariant transfer kernel learning," *IEEE Trans. Knowl. Data Eng.*, vol. 27, no. 6, pp. 1519–1532, Jun. 2015.

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参考

[1] R. J. Dolan, “情绪、认知与行为,” Sci., vol. 298, no. 5596, pp. 1191–1194, 2002. [2] R. W. Picard and R. Picard, 情感计算, vol. 252. 马萨诸塞州剑桥, USA: MIT Press, 1997. [3] C. M€uhl, B. Allison, A. Nijholt, and G. Chanel, “情感计算领域综述”

脑机接口：原理、现状与挑战,” 脑机接口, 卷 1, 第 2 期, 第 66-84 页, 2014 年。

[4] E. Lotfi 和 M.-R. Akbarzadeh-T, “实用情感神经网络,” 神经网络, vol. 59, pp. 61–72, 2014. [5] J. C. Britton, K. L. Phan, S. F. Taylor, R. C. Welsh, K. C. Berridge, 和 I. Liberzon, “社交与非社交情绪的神经关联：一项 fMRI 研究”, Neuroimage, 第 31 卷, 第 1 期, 第 397-409 页, 2006 年。

[6] A. Etkin, T. Egner, and R. Kalisch, “前额叶情绪处理”扣带皮层和内侧前额叶皮层,” 《认知科学趋势》, 第 15 卷, 第 2 期, 第 85-93 页, 2011 年。

[7] K. A. Lindquist 和 L. F. Barrett, “人脑的功能架构：情绪科学带来的新见解,” Trends Cognitive Sci., vol. 16, no. 11, pp. 533–540, 2012.

[8] R. Jenke, A. Peer, 和 M. Buss, “从脑电图识别情绪的特征提取与选择” IEEE Trans. Affective Comput., vol. 5, no. 3, pp. 327–339, 2014 年 7 月-9 月。

[9] T. Musha, Y. Terasaki, H. A. Haque, 和 G. A. Ivamitsky, “特征与情绪相关的脑电图提取,” 人工生命机器人., 第 1 卷, 第 1 期, 第 15-19 页, 1997.

[10] M.-K. Kim, M. Kim, E. Oh, 和 S.-P. Kim, “关于计算... 基于脑电信号的情绪状态估计的半监督方法,” 计算数学方法医学, 卷. 2013, 2013, 文章编号. 573734.

[11] W. 郑敏, “基于多通道脑电图的情绪识别方法——组稀疏典型相关分析,” IEEE 认知发展汇刊 Syst., vol. 9, no. 3, pp. 281–290, Sep. 2017.

[12] 李艳, 郑伟, 崔志, 宗岩, 和葛松, “基于图正则化稀疏线性回归的脑电图情绪识别,” 神经过程. 快报, 第 1-17 页, 2018 年, doi: <https://doi.org/10.1007/s11063-018-9829-1>。

[13] 郑伟立和陆北立, “研究关键频段以及基于深度神经网络的脑电图情绪识别通道,” IEEE 自动精神发展杂志, 第 7 卷, 第 3 期, 第 162-175 页, 2015 年 9 月。

[14] T.宋, W.郑, P.宋, 和 Z.崔, “脑电图情绪识别使用动态图卷积神经网络,” IEEE Affective Computing Transactions, 2018, doi: [10.1109/TAFFC.2018.2817622](https://doi.org/10.1109/TAFFC.2018.2817622).

[15] P. Pandey 和 K. Seeja, “基于脑电波的情绪状态识别”使用受试者无关方法处理信号,” 载于《数据科学与大数据分析》。纽约, 纽约州, 美国：斯普林格出版社, 2019 年, 第 117-124 页。

[16] 李翔, 宋东, 张鹏, 张宇, 侯宇, 和 胡博, “探索跨被试情绪识别中的脑电图特征,” 神经科学前沿, 第 12 卷, 2018 年, 文章编号 162。

[17] Y. Ganin, E. Ustinova, H. Ajakan, P. Germain, H. Larochelle, F. Laviolette、M. Marchand 和 V. Lempitsky, “神经网络的领域对抗训练,” 机器学习研究杂志, 第 17 卷, 第 59 期, 第 1-35 页, 2016.

[18] 李艳, 郑伟, 崔志, 张涛, 宗彦, “一种新型神经基于大脑半球不对称的脑电情感识别网络模型,” 在 2018 年第 27 届国际人工智能联合会议上, 第 1561-1567 页。

[19] D. M. Tucker, 《大脑的侧向功能、情绪与概念化》, 《心理季刊》, 第 89 卷, 第 1 期, 1981 年, 文章编号 19。 [20] N. Geschwind 和 A. M. Galaburda, 《大脑的偏侧化：生物学机制、关联和病理学： I. 一个假设和研究的计划,” 神经病学档案, 卷 42, 第 5 期, 第 428-459 页, 1985 年。

[21] R. J. Zatorre, M. Jones-Gotman, A. C. Evans, and E. Meyer, “人类嗅觉皮层的功能定位和偏侧化,” Nature, 卷 360, 第 6402 期, 第 339–340 页, 1992 年。

授权使用受限于：福建师范大学。于 2025 年 9 月 26 日 08:33:38 UTC 从 IEEE Xplore 下载。适用限制。

[22] D. N. Greve, L. Van der Haegen, Q. Cai, S. Stufflebeam, M. R. Sabuncu、B. Fischl 和 M. Brysbaert, “基于表面的语言偏侧化和皮质不对称性分析,” 认知神经科学杂志, 第 25 卷, 第 9 期, 第 1477-1492 页, 2013 年。

[23] S. J. Dimond、L. Farrington 和 P. Johnson, “不同的情感来自左右半球的响应,” Nature, 第 261 卷, 第 5562 期, 第 690-692 页, 1976 年。

[24] R. J. Davidson, P. Ekman, C. D. Saron, J. A. Senulis, 和 W. V. Friesen, “接近-退缩与大脑不对称性：情感表达与大脑生理学： I,” J. Personality Social Psychology, 卷. 58, 第 2 期, 1990, 文章编号. 330.

[25] R. J. Davison, “大脑不对称性、情绪和情感风格,” 在《大脑不对称性》中, R. J. Davidson 和 K. Hugdahl 编辑, 马萨诸塞州剑桥, 美国: 麻省理工学院出版社, 1995, 第 361-387 页。

[26] R. J. Davidson, D. Pizzagalli, J. B. Nitschke, 和 K. Putnam, “抑郁症：来自情感神经科学的视角,” 年度评论心理学, 第 53 卷, 第 1 期, 第 545-574 页, 2002 年。

[27] E. Y. Costanzo, M. Villarreal, L. J. Drucaroff, M. Ortiz-Villafañe, M. N. Castro, M. Goldschmidt, A. E. Wainsztein, M. S. Ladrón-de Guevara, C. Romero, L. I. Brusco, 等, "情感反应的半球专业化、语言的大脑优势半球和左右利手：情绪、语言和灵巧性的偏侧化", 《行为脑研究》, 第 288 卷, 第 11-19 页, 2015 年。

[28] M. J. van Putten, “The revised brain symmetry index,” 临床神经生理学, 第 118 卷, 第 11 期, pp. 2362–2367, 2007. [29] H. Hinrikus, A. Suhhova, M. Bachmann, K. Adamsoo、€U. V~ohma, J. Lass, 和 V. Tuulik, “用于检测抑郁症的脑电图频谱不对称指数,” 医学生物工程与计算, 第 47 卷, 第 12 期, 2009, 第 1291 号。

[30] J. M. Peters, M. Tomas-Fernandez, M. J. van Putten, 和 T. Loddenkemper, “在儿童 WADA 测试中使用脑对称指数进行行为测量和脑电图监测,” 癫痫行为学, 第 23 卷, 第 3 期, pp. 247–253, 2012.

[31] S. Hochreiter 和 J. Schmidhuber, “长短期记忆,” 神经计算, 第 9 卷, 第 8 期, 第 1735-1780 页, 1997 年。

[32] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville 和 Y. Bengio, “生成对抗网络,” 在《神经信息处理系统会议录》2014 年, 第 2672–2680 页。

[33] A. Jain, A. Singh, H. S. Koppula, S. Soh 和 A. Saxena, “循环神经网络通过感官融合架构进行驾驶员活动预期,” 在 2016 年 IEEE 国际机器人与自动化会议论文集中, 第 3118-3125 页。

[34] 王伟, 崔志强, 闫勇, 冯静, 闫松, 舒翔, 和 Sebe·N, “循环面部老化,” 在 2016 年 IEEE 计算机视觉与模式识别会议论文集中, 第 2378-2386 页。

[35] L. Bottou, “大规模机器学习与随机梯度下降,” 在 2010 年 COMPSTAT 会议论文集中, 第 177-186 页。 [36] 郑伟立和卢伯伦, “基于迁移学习的个性化脑电图情感模型,” 在 2016 年第 25 届国际联合人工智能会议论文集中

Intell., 2016, pp. 2732–2738.

[37] J. A. Suykens and J. Vandewalle, “最小二乘支持向量机分类器,” 神经处理快报, vol. 9, no. 3, pp. 293–300, 1999. [38] L. Breiman, “随机森林,” 机器学习, vol. 45, no. 1, pp. 5–32, 2001.

[39] B. Thompson, “典型相关分析,” 统计行为科学百科全书, Wiley 在线图书馆, 2005. [40] M. Defferrard, X. Bresson, and P. Vandergheynst, “图上的卷积神经网络与快速局部谱过滤,” 在 2016 年神经信息处理系统会议论文集中, pp. 3844–3852.

[41] M. Sugiyama, S. Nakajima, H. Kashima, P. V. Buenau, and M. Kawanabe, “直接重要性估计与模型选择及其在协变量偏移适应中的应用”, 在《会议论文集》中。神经信息处理系统, 2008, 第 1433-1440 页。

[42] T. Kanamori, S. Hido, 和 M. Sugiyama, “一种最小二乘法”一种直接重要性估计方法,” J. Mach. Learn. Res., vol. 10, pp. 1391–1445, 2009.

[43] W.-S. Chu, F. De la Torre, 和 J. F. Cohn, “用于个性化面部表情分析的迁移机器,” IEEE Trans. 模式分析、机器智能, 第 39 卷, 第 3 期, 第 529-545 页, 2017 年 3 月。

[44] B. Scholkopf, A. Smola, and K.-R. Muller, “Nonlinear component 分析作为核特征值问题,” 神经计算, vol. 10, no. 5, pp. 1299–1319, 1998.

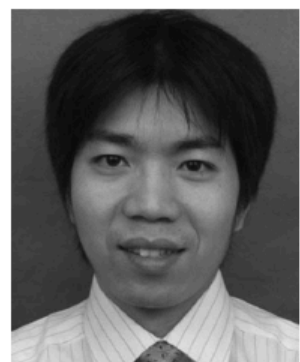
[45] S. J. Pan, I. W. Tsang, J. T. Kwok 和 Q. Yang, “领域自适应通过迁移成分分析,” IEEE 神经网络汇刊, 第 22 卷, 第 2 期, 第 199-210 页, 2011 年。

[46] M. Long, J. Wang, J. Sun, 和 S. Y. Philip, “域不变迁移核学习,” IEEE 知识数据工程汇刊, 第 27 卷, 第 6 期, 第 1519-1532 页, 2015 年 6 月。

- [47] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 2960–2967.
- [48] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 2066–2073.
- [49] R. Collobert, F. Sinz, J. Weston, and L. Bottou, "Large scale transductive svms," *J. Mach. Learn. Res.*, vol. 7, no. Aug., pp. 1687–1712, 2006.
- [50] E. Sangineto, G. Zen, E. Ricci, and N. Sebe, "We are not all equal: Personalizing models for facial expression analysis with transductive parameter transfer," in *Proc. 22nd ACM Int. Conf. Multimedia*, 2014, pp. 357–366.
- [51] H. Becker, J. Fleureau, P. Guillotel, F. Wendling, I. Merlet, and L. Albera, "Emotion recognition based on high-resolution eeg recordings and reconstructed brain sources," *IEEE Trans. Affective Comput.*, 2017, doi: [10.1109/TAFFC.2017.2768030](https://doi.org/10.1109/TAFFC.2017.2768030).
- [52] J. Donahue, Y. Jia, O. Vinyals, J. Hoffman, N. Zhang, E. Tzeng, and T. Darrell, "Decaf: A deep convolutional activation feature for generic visual recognition," in *Proc. Int. Conf. Mach. Learn.*, 2014, pp. 647–655.



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[47] B. Fernando, A. Habrard, M. Sebban, 和 T. Tuytelaars, “基于子空间对齐的无监督视觉域适应,” IEEE 国际计算机视觉会议论文集, 2013 年, 第 2960-2967 页。

[48] Gong B., Shi Y., Sha F. 和 Grauman K., “Geodesic flow kernel for unsupervised domain adaptation,” in Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2012, pp. 2066–2073.

[49] R. Collobert、F. Sinz、J. Weston 和 L. Bottou, “大规模“转导式支持向量机,” J. Mach. Learn. Res., vol. 7, no. Aug., pp. 1687–1712, 2006.

[50] E. Sangineto, G. Zen, E. Ricci, and N. Sebe, “我们并不都平等: 将模型个性化用于面部表情分析, 通过转导参数迁移,” 在第二十二届 ACM 国际会议论文集中 多媒体, 2014, 第 357-366 页。

[51] H. Becker, J. Fleureau, P. Guillotel, F. Wendling, I. Merlet, 和 L. Albera, “基于高分辨率脑电图记录和重建脑源的情绪识别,” IEEE 情感计算汇刊, 2017, DOI: 10.1109/TAFFC.2017.2768030.

[52] J. Donahue, Y. Jia, O. Vinyals, J. Hoffman, N. Zhang, E. Tzeng, 和 T. Darrell, “Decaf: 一种用于通用视觉识别的深度卷积激活特征”, 在 2014 年国际机器学习会议论文集中, 第 647-655 页。



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